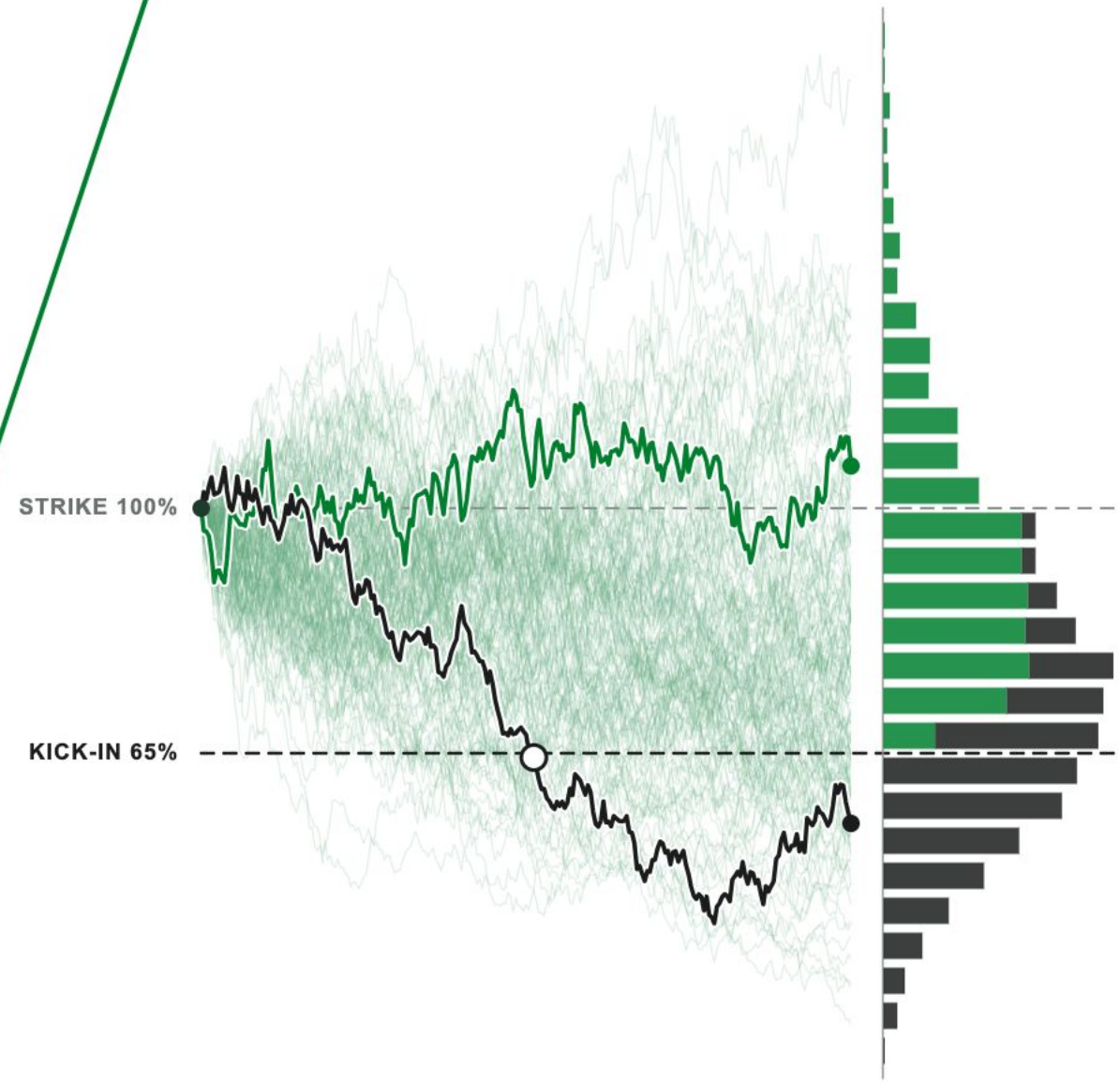
UNIVERSITY OF ST.GALLEN · CORPORATE FINANCE (7,107)

Monte Carlo Option Pricing

Pricing a Worst-Of Multi Barrier Reverse Convertible on Adobe, Apple and Microsoft

Group III · Ali Bozan Can, Benjamin Marco Bolliger, Cédric Anwar El Soda, Ivan Kissinger, Joaquim Graber, Leandro Broccucci, Mirko Cirantineo



The starting point: 1000$ to invest for an expected sideways or slightly down year

Daily closing prices of Adobe, Apple and Microsoft from 10/25 to 10/26



Sources: Bloomberg (2026); own Excel model, sheet 3

The starting point: 1000$ to invest for an expected sideways or slightly down year

Daily closing prices of Adobe, Apple and Microsoft from 10/25 to 10/26



USD 1,000 · 1 year

Consensus: 86% of analysts issue a HOLD rating for the three underlyings; no growth catalyst ahead and downside risks already priced in

Sources: Bloomberg (2026); own Excel model, sheet 3

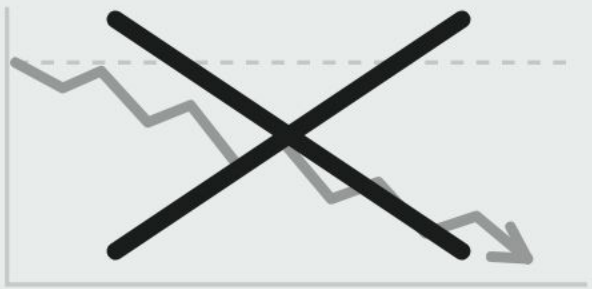
Which product expresses the market view adequately?

Three ways to invest USD 1,000 for one year · Consensus: HOLD

Sources: Bloomberg (2026) (USD 1-year rate); own Excel model, sheet 2; own illustration

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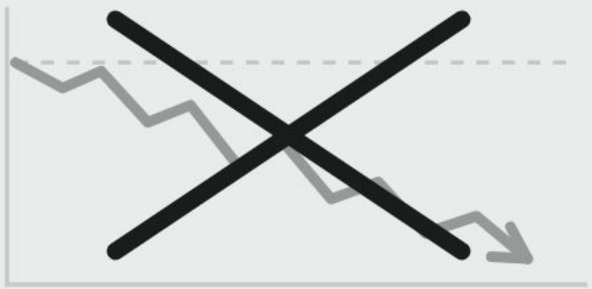
~~Shares~~

You are exposed to the full downside

Sources: Bloomberg (2026) (USD 1-year rate); own Excel model, sheet 2; own illustration

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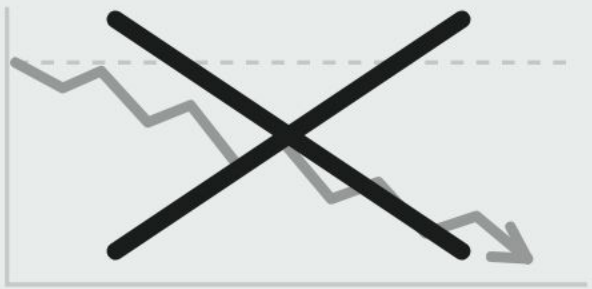
~~Government bond~~

Safe, but only ≈ 4.5% a year

Sources: Bloomberg (2026) (USD 1-year rate); own Excel model, sheet 2; own illustration

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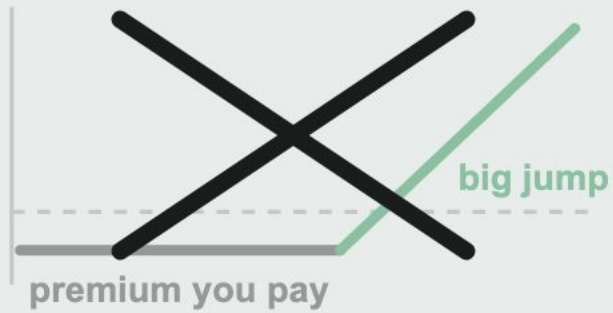
~~Shares~~

You are exposed to the full downside



~~Government bond~~

Safe, but only ≈ 4.5% a year



~~Call option~~

You are not expecting an increase in the share price

None are suitable for a sideways or slightly downward market

Sources: Bloomberg (2026) (USD 1-year rate); own Excel model, sheet 2; own illustration

Banks' solution: A Worst-of Multi-BRC

Banks created a product for this exact market view

The product: A worst-of Multi-BRC on $ADBE, $AAPL, $MSFT

NOMINAL
USD 1,000
(=Issue price)

MATURITY
1 year
Initial fixing date: 6/10/26
Expiration: 6/10/27

KICK-IN BARRIER
65%
of underlying's reference price

UNDERLYINGS
Worst of 3
- Adobe
- Apple
- Microsoft

ISSUER
UBS
simulated with UBS as the issuer

15% p.a.

SEMI-ANNUAL COUPON
2 × 75 per 1,000 USD, paid whatever happens, but never more

Sources: UBS AG (2018) (model for the structure); Bloomberg (2026); own Excel model, sheets 1–2

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But is it fair for the risk you take on?

Sources: UBS AG (2018) (model for the structure); Bloomberg (2026); own Excel model, sheets 1–2

What is Monte Carlo?

At its core, Monte Carlo computes expected values

1946

Stanislaw Ulam, recovering from an illness, plays solitaire: why not play **100 games** and **count the wins**?

EXPECTED VALUE

x_i	
$P(x_i)$	$\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$

$$\mathrm{E}(X) = \sum x_i \cdot P(x_i) = (1+2+3+4+5+6) \cdot \frac{1}{6} = \frac{21}{6} = 3.5$$

Sources: Eckhardt (1987), p. 131; Metropolis (1987), p. 127; Haigh et al. (2014); dice example adapted from The Synthetic Mind (2026), about 1:20; own simulation

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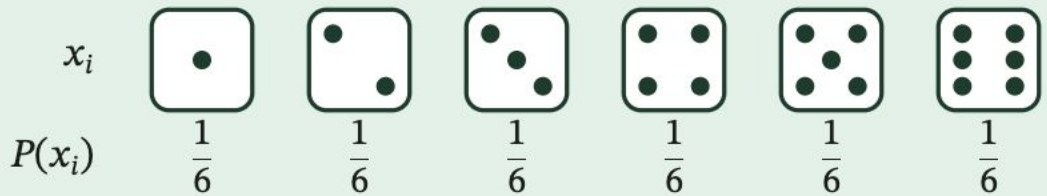
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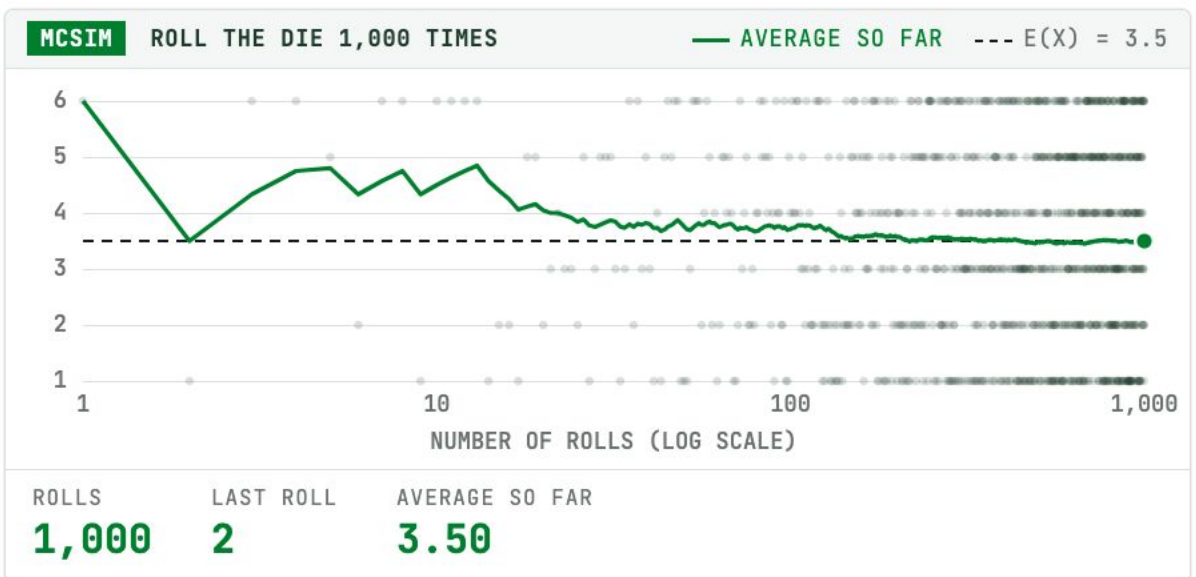
1947

With John von Neumann: neutrons in atomic bombs, for Los Alamos

EXPECTED VALUE



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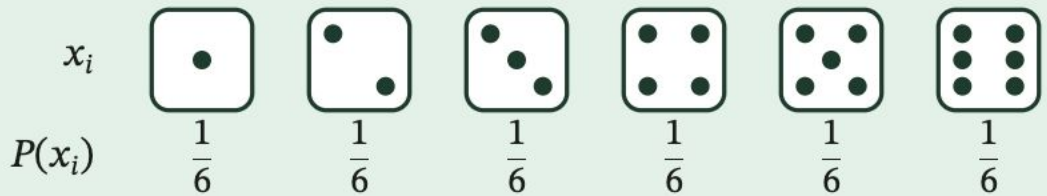
With John von Neumann: neutrons in atomic bombs, for Los Alamos

The name

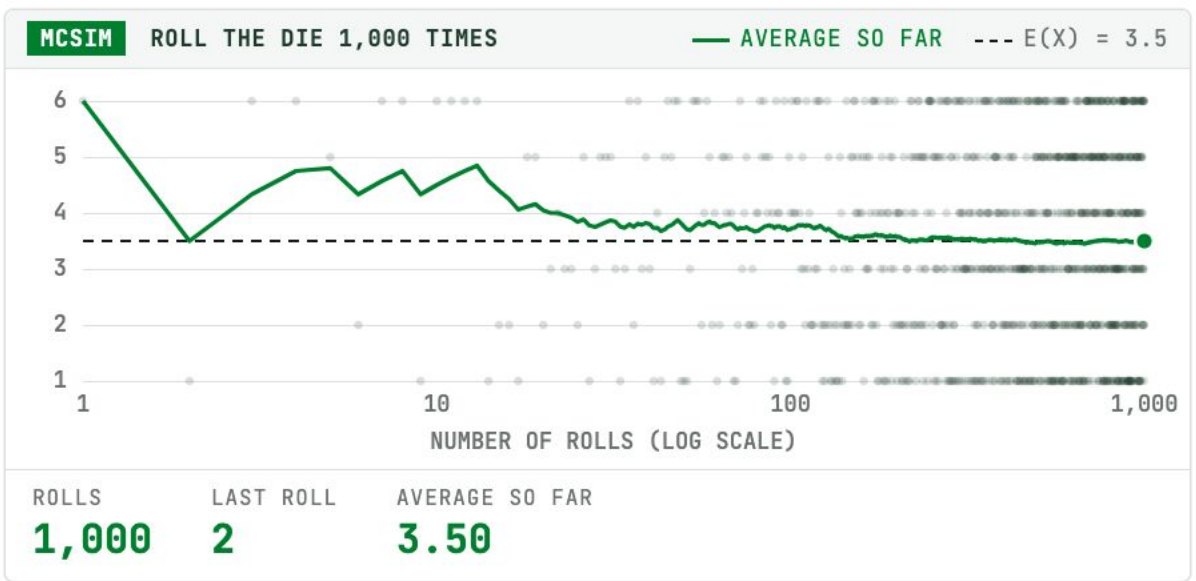
Their colleague Nicholas Metropolis suggests it: Ulam had an uncle who "just had to go to Monte Carlo"

Metropolis (1987), p. 127

EXPECTED VALUE



$$\mathrm{E}(X) = \sum x_i \cdot P(x_i) = (1+2+3+4+5+6) \cdot \frac{1}{6} = \frac{21}{6} = 3.5$$



average of 1,000 rolls

≈ 3.5

for a continuous variable:

$$\mathrm{E}(X) = \int x \cdot f(x)\, dx$$

Sources: Eckhardt (1987), p. 131; Metropolis (1987), p. 127; Haigh et al. (2014); dice example adapted from The Synthetic Mind (2026), about 1:20; own simulation

Monte Carlo: averaging the payoffs of random paths

The price is a discounted expected payoff, estimated by averaging simulated years

LECTURE 4: RISK-NEUTRAL VALUATION

$$\text{price} = \underbrace{e^{-rT}}_{\text{discounted at the risk-free rate}} \cdot \underbrace{\mathrm{E}^{Q}[\text{payoff}]}_{\text{expected payoff in a risk-neutral world}}$$

lecture: r_f

Sources: Arnold (2026), slides 11 and 15; explanation adapted from The Synthetic Mind (2026); own simulation

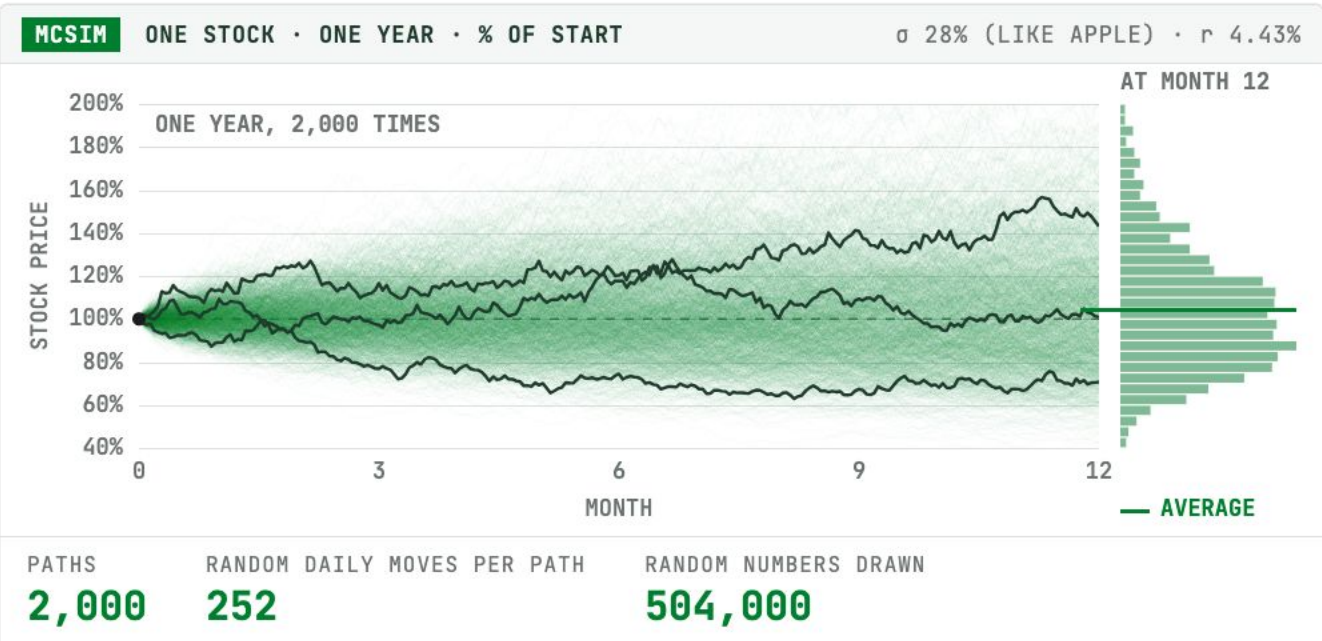
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Our payoff depends on the **whole year**, so its expectation is an integral over **252 random daily moves**, one for each trading day.

Sources: Arnold (2026), slides 11 and 15; explanation adapted from The Synthetic Mind (2026); own simulation

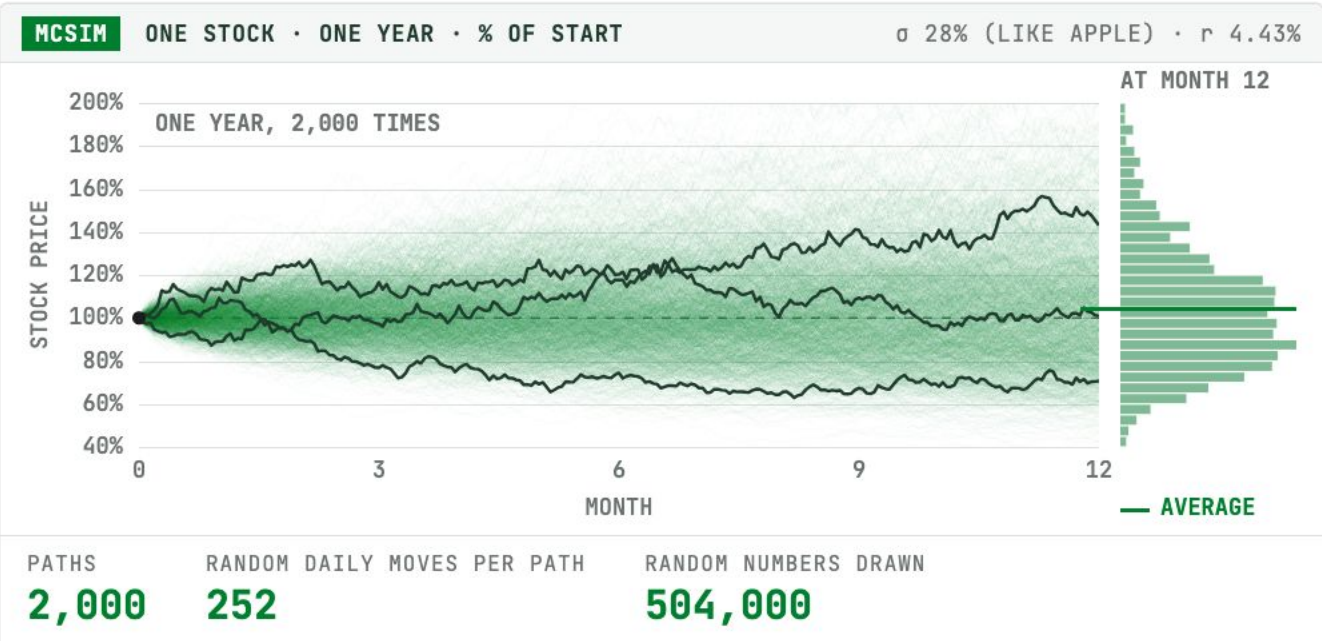
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Sample a random path → **Compute its discounted payoff** → **Average them all**

LAW OF LARGE NUMBERS

$$\frac{1}{n}\sum_{i=1}^{n} \text{payoff}_i \overset{n \to \infty}{\longrightarrow} \mathrm{E}^{Q}[\text{payoff}]$$

the average converges to the true value

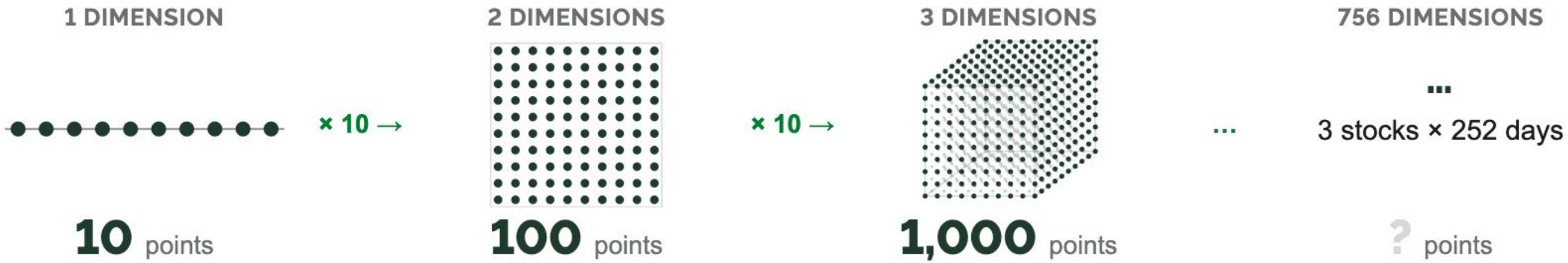
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Monte Carlo escapes the curse of dimensionality

A grid's points explode with every dimension. Random points don't.

One random number is one dimension. A simulated year of three stocks needs 756 of them, so it is **a single point in 756-dimensional space**.

A GRID: THE POINTS MULTIPLY WITH EVERY DIMENSION



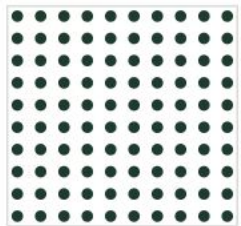
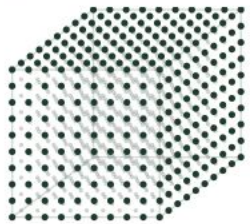
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A GRID: THE POINTS MULTIPLY WITH EVERY DIMENSION

1 DIMENSION	2 DIMENSIONS	3 DIMENSIONS	756 DIMENSIONS
			...
× 10 →	× 10 →	...	3 stocks × 252 days
			ten values for each
10 points	**100** points	**1,000** points	10^{756} points

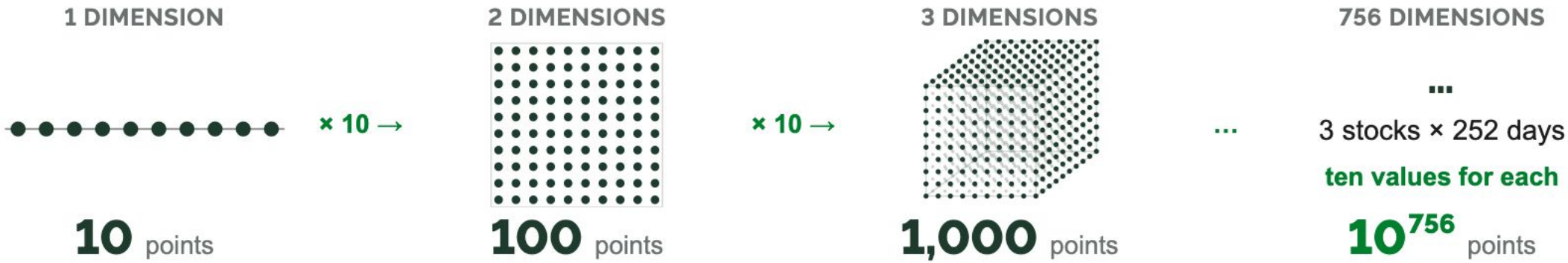
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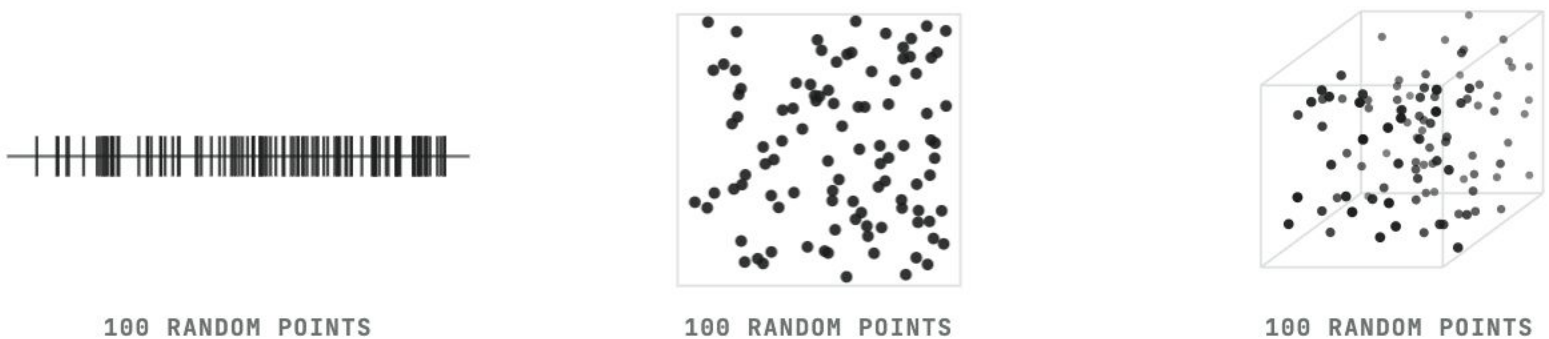
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A GRID: THE POINTS MULTIPLY WITH EVERY DIMENSION



MCSIM RANDOM POINTS · THE SAME n IN ANY DIMENSION



756 dimensions: still 100 points

Monte Carlo:

$$\text{error} \propto \frac{1}{\sqrt{n}}$$

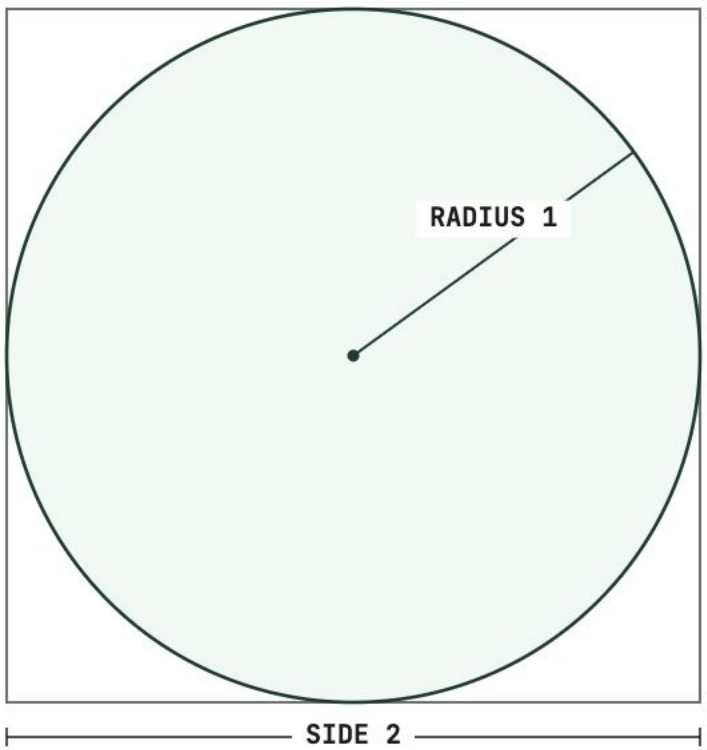
in any dimension

Sources: Glasserman (2003), ch. 1; adapted from The Synthetic Mind (2026); own illustration

How precise is Monte Carlo?

Because π is already known, we can measure the error exactly.

MCSIM RANDOM DARTS ON THE SQUARE • INSIDE • OUTSIDE



DARTS	INSIDE	π ≈ 4 × INSIDE / DARTS	95% INTERVAL
0	0	–	–

SHARE OF THE AREA

$$\text{share inside} = \frac{\pi \cdot 1^2}{2 \cdot 2} = \frac{\pi}{4}$$

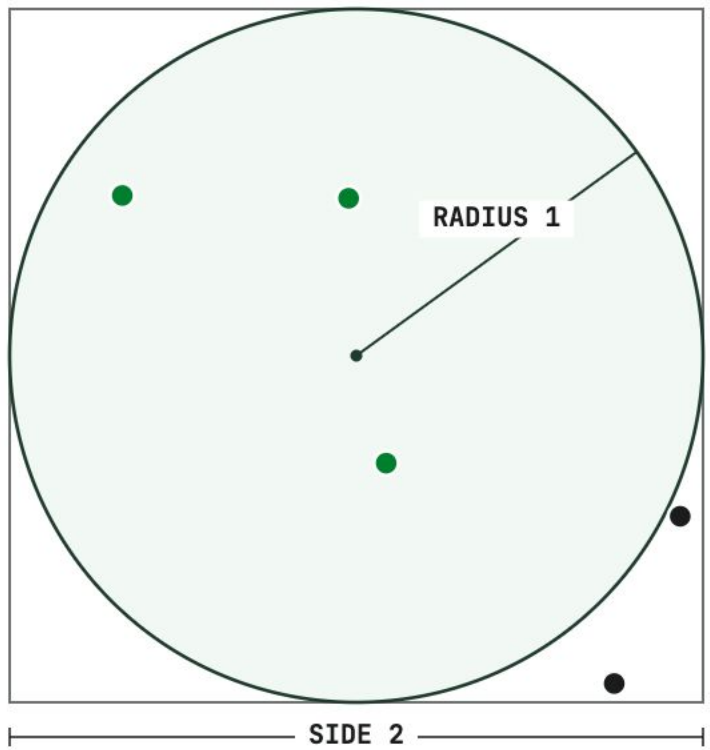
$$\pi \approx 4 \times \frac{\text{darts inside}}{\text{darts thrown}}$$

Sources: own simulation; adapted from The Synthetic Mind (2026), 3:00–5:12

How precise is Monte Carlo?

Because π is already known, we can measure the error exactly.

MCSIM RANDOM DARTS ON THE SQUARE • INSIDE • OUTSIDE



DARTS	INSIDE	π ≈ 4 × INSIDE / DARTS	95% INTERVAL
5	3	2.40	–

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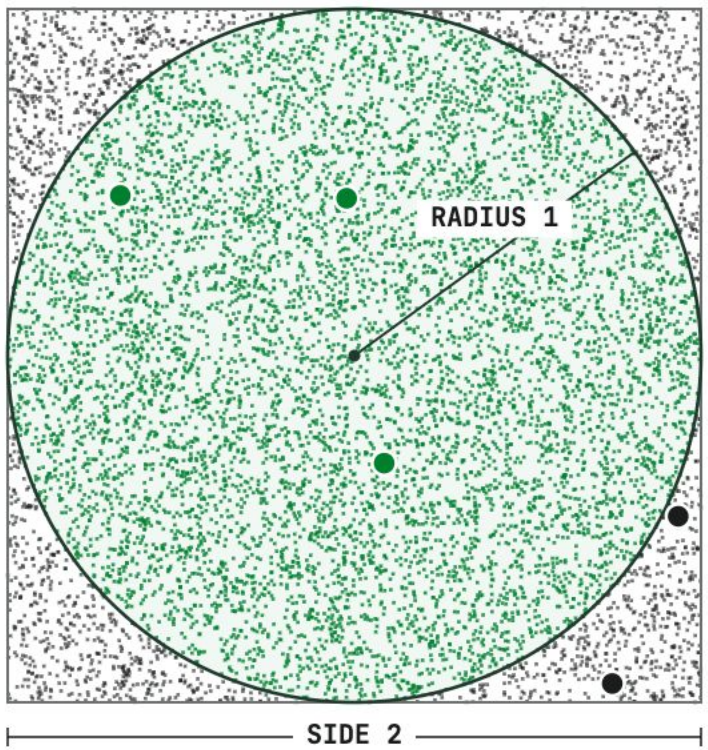
5 darts	3 inside	4 × 3/5 = 2.40

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Because π is already known, we can measure the error exactly.

MCSIM RANDOM DARTS ON THE SQUARE • INSIDE • OUTSIDE



DARTS	INSIDE	π ≈ 4 × INSIDE / DARTS	95% INTERVAL
10,000	7,875	3.150	3.12–3.18

SHARE OF THE AREA

$$\text{share inside} = \frac{\pi \cdot 1^2}{2 \cdot 2} = \frac{\pi}{4}$$

$$\pi \approx 4 \times \frac{\text{darts inside}}{\text{darts thrown}}$$

5 darts	3 inside	4 × 3/5 = 2.40
10,000 darts	7,875 inside	4 × 7,875/10,000 = 3.150

95% CONFIDENCE INTERVAL

$$3.150 \pm 1.96 \times 0.0164 = 3.12 \text{ to } 3.18$$

standard error $4\sqrt{p(1-p)/n} = 0.0164$ with $p = 7{,}875 / 10{,}000$

error $\propto 1/\sqrt{n}$: half the error takes 4× the darts. That's why our price comes with an interval, too.

Sources: own simulation; adapted from The Synthetic Mind (2026), 3:00–5:12

Geometric Brownian motion: building one simulated year

252 daily moves on each of three stocks, 756 random numbers per path

Lecture reference: **Lecture 4, slide 15**

1977

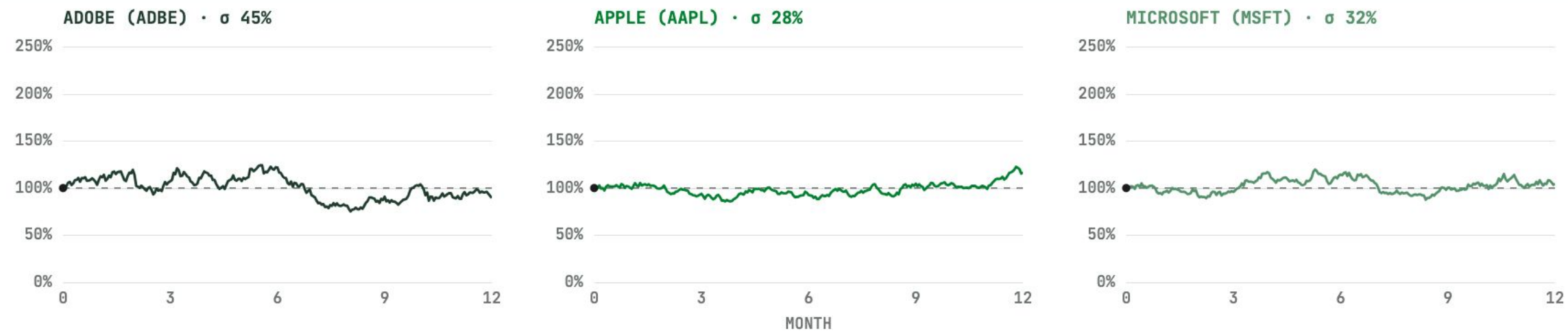
Phelim Boyle brings Monte Carlo to option pricing

Each dart becomes a whole year:

1 path = 252 random daily moves × 3 stocks

GBM THREE STOCKS AT ONCE · DAILY STEPS · % OF START

OUR EXCEL'S INPUTS



PATHS (SIMULATED YEARS)	RANDOM NUMBERS PER PATH · 252 DAYS × 3 STOCKS	RANDOM NUMBERS DRAWN
1	756	756

Sources: Boyle (1977); Arnold (2026), slide 15; own simulation

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GEOMETRIC BROWNIAN MOTION

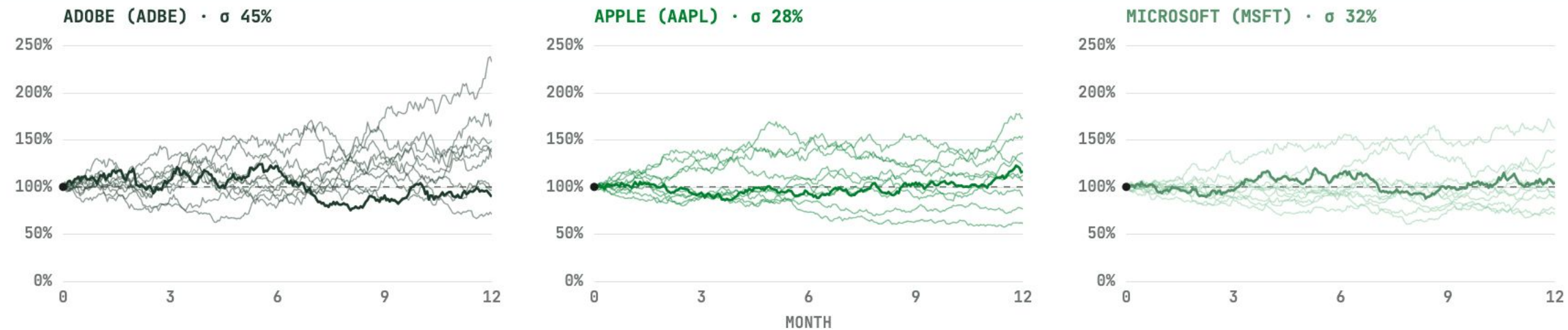
$$dS_t = (r - q) \cdot S_t dt + \sigma \cdot S_t dW_t$$

the lecture's model for a stock price

lecture: r_f and δ

GBM THREE STOCKS AT ONCE · DAILY STEPS · % OF START

OUR EXCEL'S INPUTS



PATHS (SIMULATED YEARS)	RANDOM NUMBERS PER PATH · 252 DAYS × 3 STOCKS	RANDOM NUMBERS DRAWN
10	756	7,560

Sources: Boyle (1977); Arnold (2026), slide 15; own simulation

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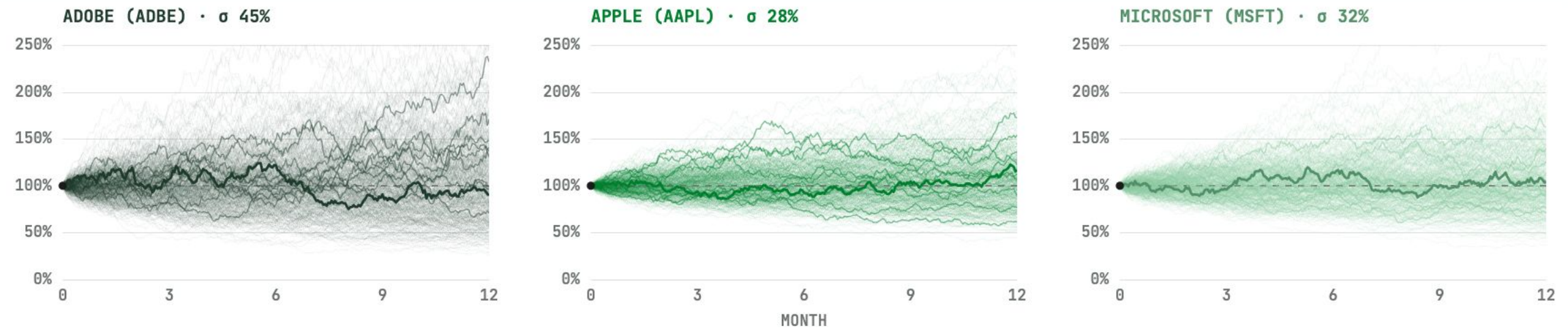
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OUR EXCEL'S INPUTS



PATHS (SIMULATED YEARS)	RANDOM NUMBERS PER PATH · 252 DAYS × 3 STOCKS	RANDOM NUMBERS DRAWN
500	756	378,000

Sources: Boyle (1977); Arnold (2026), slide 15; own simulation

Worst-Of Multi-Barrier Reverse Convertible

BRCs are part of Yield Enhancement products

Investment Products

11
Capital protection

1100
1130
1135
1140

12 Yield enhancement

1200	Discount Certificate
1210	Barrier Discount Certificate
1220	Reverse Convertible
1230	Barrier Reverse Convertible
1255	Conditional Coupon Reverse Convertible
1260	Conditional Coupon Barrier Reverse Convertible

13
Participation

1300
1310
1320
1330
1340

14
Investment products with additional credit risk

1400
1410
1420
1430

Leverage products

20
Leverage

2100
2110
2200
2210
2300

Sources: Swiss Structured Products Association (2026); UBS AG (2018); Bloomberg (2026); own Excel model, sheets 1–2; own illustration

Worst-Of Multi-Barrier Reverse Convertible

BRCs are part of Yield Enhancement products

Investment Products

11 Capital protection

1100
1130
1135
1140

12 Yield enhancement

You give up upside for a high, fixed coupon

1200	Discount Certificate
1210	Barrier Discount Certificate
1220	Reverse Convertible
1230	Barrier Reverse Convertible
1255	Conditional Coupon Reverse Convertible
1260	Conditional Coupon Barrier Reverse Convertible

13 Participation

1300
1310
1320
1330
1340

14 Investment products with additional credit risk

1400
1410
1420
1430

Leverage products

20 Leverage

2100
2110
2200
2210
2300

Sources: Swiss Structured Products Association (2026); UBS AG (2018); Bloomberg (2026); own Excel model, sheets 1–2; own illustration

Worst-Of Multi-Barrier Reverse Convertible

BRCs are part of Yield Enhancement products

Investment Products | Leverage products

11 Capital protection

1100
1130
1135
1140

12 Yield enhancement

You give up upside for a high, fixed coupon

1200 Discount Certificate
1210 Barrier Discount Certificate
1220 Reverse Convertible
1230 Barrier Reverse Convertible
1255 Conditional Coupon Reverse Convertible
1260 Conditional Coupon Barrier Reverse Convertible

13 Participation

1300
1310
1320
1330
1340

14 Investment products with additional credit risk

1400
1410
1420
1430

20 Leverage

2100
2110
2200
2210
2300

Worst-of multi barrier reverse convertible (Multi-BRC)

NOMINAL
USD 1,000
(=Issue price)

MATURITY
1 year
Initial fixing date: 6/10/26
Expiration: 6/10/27

KICK-IN BARRIER
65%
of underlying's reference price

UNDERLYINGS
Worst of 3
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- Apple
- Microsoft

ISSUER
UBS
simulated with UBS as the issuer

15% p.a.

SEMI-ANNUAL COUPON
2 × 75 per 1,000 USD

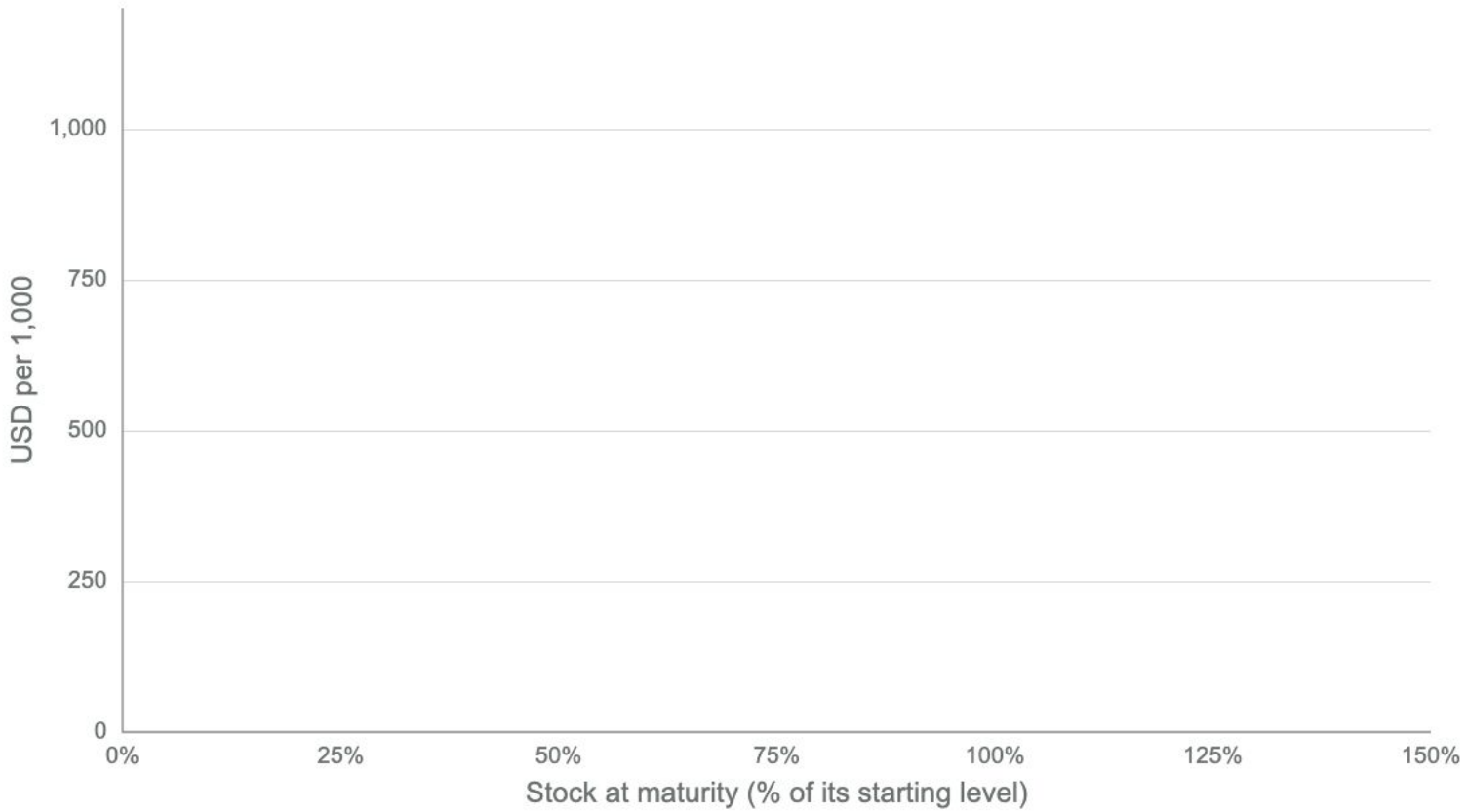
RISK-FREE RATE
4.43% continuously compounded ≈ 4.5% a year

Sources: Swiss Structured Products Association (2026); UBS AG (2018); Bloomberg (2026); own Excel model, sheets 1–2; own illustration

The Multi-BRC Explained

From a BRC to a Worst-Of Multi-BRC

Repayment at maturity, USD per 1,000 · plus USD 150 in coupons in every case



1

2

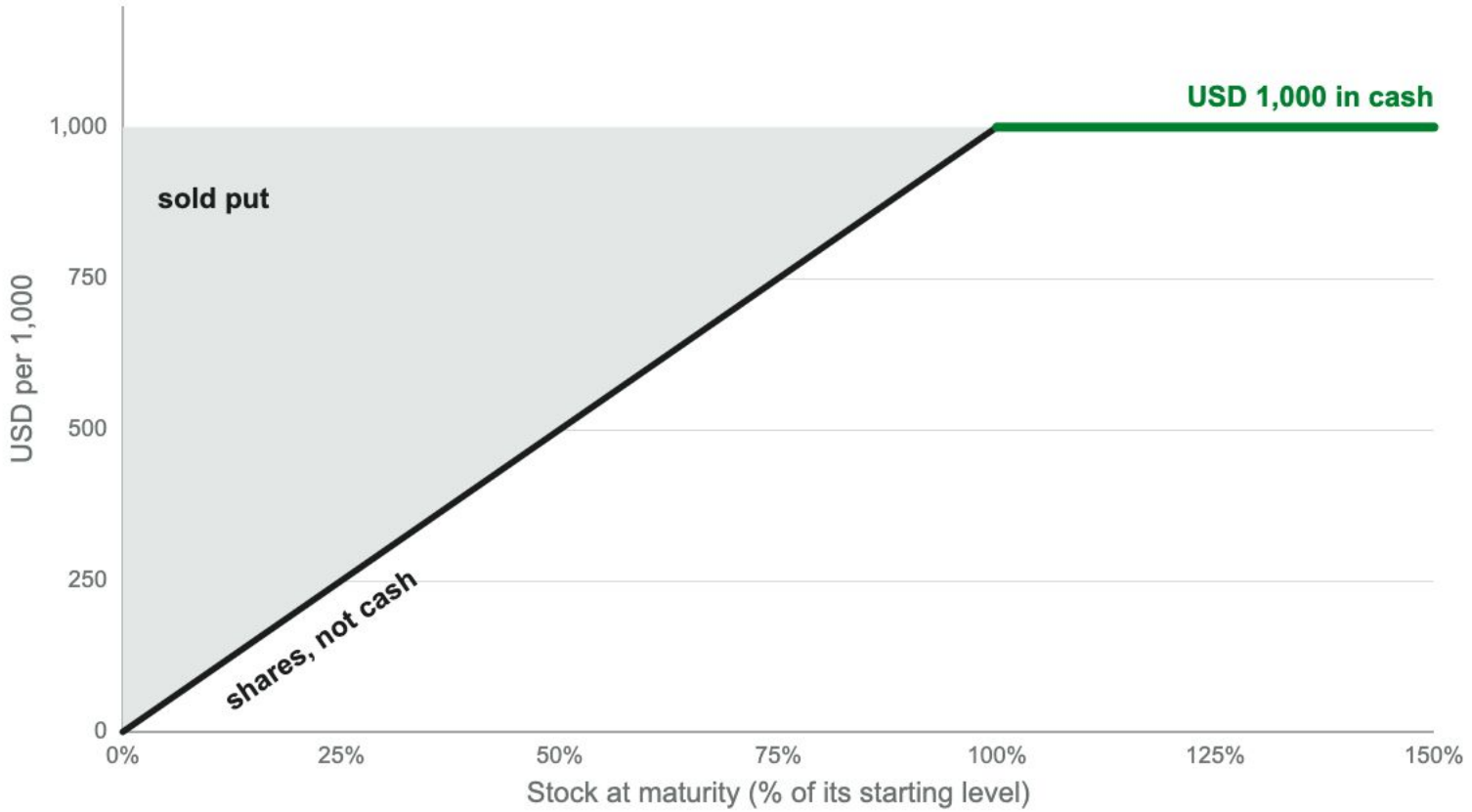
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Sources: Arnold (2026), slide 14; adapted from UBS AG (2018); own illustration

The Multi-BRC Explained

From a BRC to a Worst-Of Multi-BRC

Repayment at maturity, USD per 1,000 · plus USD 150 in coupons in every case



1 Reverse convertible

= bond + sold put

Below strike (100%): You receive shares

2

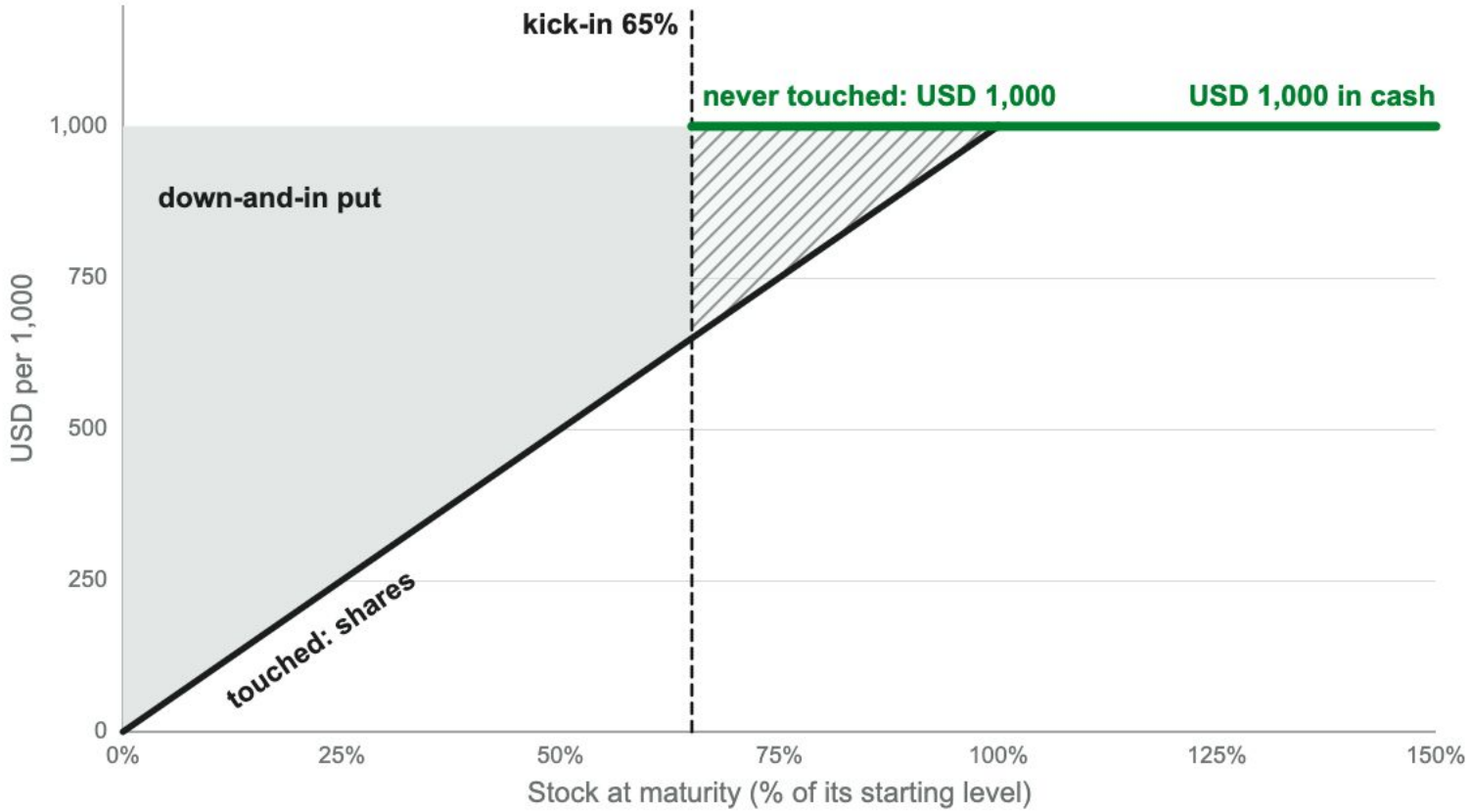
3

Sources: Arnold (2026), slide 14; adapted from UBS AG (2018); own illustration

The Multi-BRC Explained

From a BRC to a Worst-Of Multi-BRC

Repayment at maturity, USD per 1,000 · plus USD 150 in coupons in every case



Reverse convertible

1 **= bond + sold put**

Below strike (100%): You receive shares

+ Barrier at 65%

2 **= down-and-in put**

Touch of the barrier turns the product into a reverse convertible

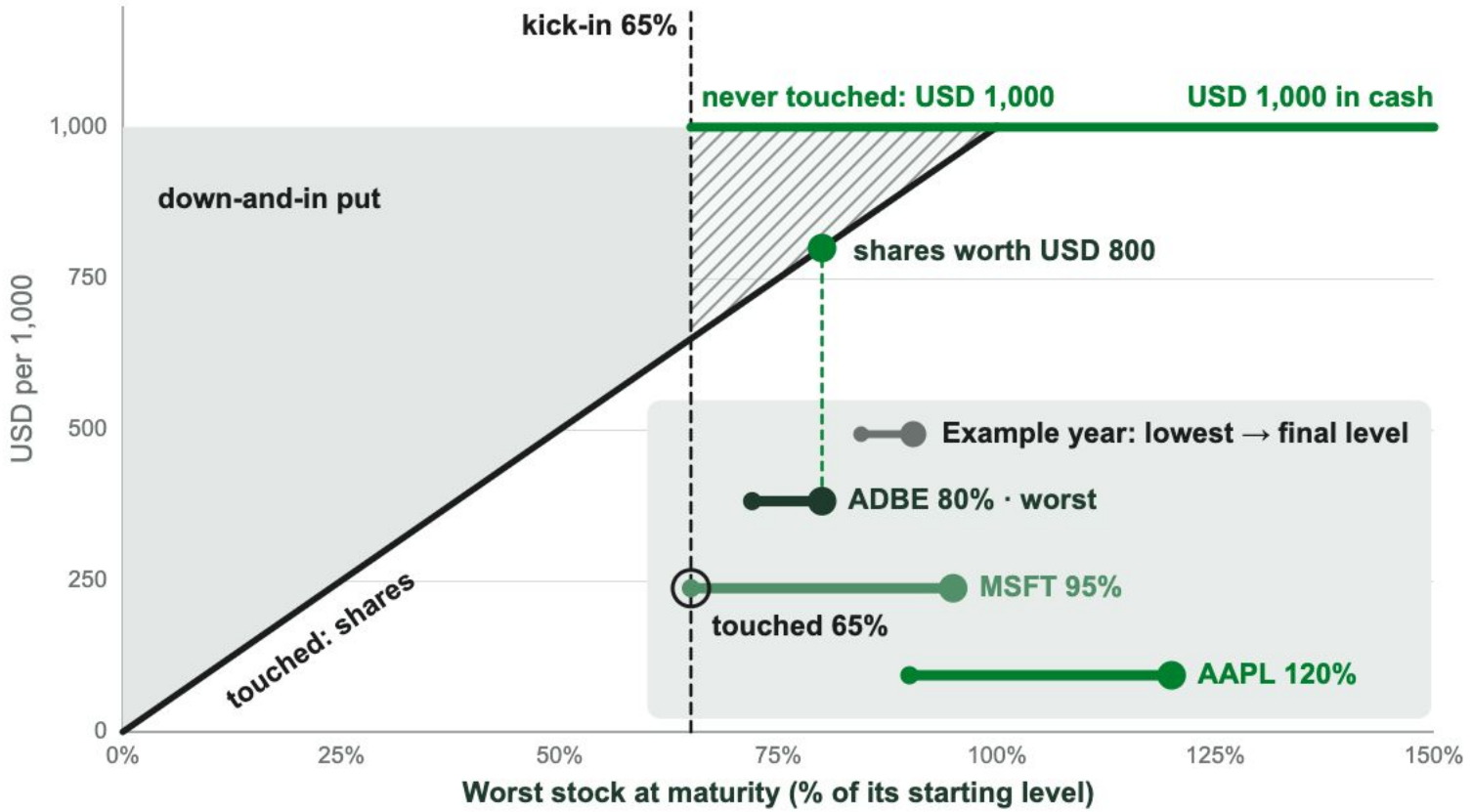
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Sources: Arnold (2026), slide 14; adapted from UBS AG (2018); own illustration

The Multi-BRC Explained

From a BRC to a Worst-Of Multi-BRC

Repayment at maturity, USD per 1,000 · plus USD 150 in coupons in every case



1 **Reverse convertible**
= bond + sold put
Below strike (100%): You receive shares

2 **+ Barrier at 65%**
= down-and-in put
Touch of the barrier turns the product into a reverse convertible

3 **+ Worst Of Principle**
underlying asset with the worst performance counts

Sources: Arnold (2026), slide 14; adapted from UBS AG (2018); own illustration

Payoff scenarios per 1,000 invested

You always get the coupons, then either your 1,000 back or shares of the worst stock

COUPONS

150

= 2 × 75

USD per 1,000

- **75** after 6 months
- **75** after 12 months

always paid

Sources: adapted from UBS AG (2018); own illustration

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always paid

? **Kick-in: did any stock touch 65%?** → **No** → 1 **USD 1,000 in cash**

Sources: adapted from UBS AG (2018); own illustration

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always paid

? **Kick-in: did any stock touch 65%?**

No → **1** **USD 1,000 in cash**

Yes

? **Is every stock at or above its strike, its starting price?**

Yes → **2** **USD 1,000 in cash**

Sources: adapted from UBS AG (2018); own illustration

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You always get the coupons, then either your 1,000 back or shares of the worst stock

COUPONS

150

= 2 × 75

USD per 1,000

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- **75** after 12 months

always paid

? **Kick-in: did any stock touch 65%?**

No → **1 USD 1,000 in cash**

Yes

? **Is every stock at or above its strike, its starting price?**

Yes → **2 USD 1,000 in cash**

No

3 Shares of the worst stock

1,000 ÷ its starting price; fractions in cash

Sources: adapted from UBS AG (2018); own illustration

Payoff scenarios per 1,000 invested

You always get the coupons, then either your 1,000 back or shares of the worst stock

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No → **1 USD 1,000 in cash**

Yes

? **Is every stock at or above its strike, its starting price?**

Yes → **2 USD 1,000 in cash**

No

3 Shares of the worst stock

1,000 ÷ its starting price; fractions in cash

Example: worst stock −20%

Shares worth 1,000 × 80%	**800**
Coupons 2 × 75	**150**
You get	**USD 950**

Sources: adapted from UBS AG (2018); own illustration

When it suits you, and the risks

Best in a sideways or slightly-down year

Advantages

Fixed coupon, generally above the market interest rate

You get a fixed coupon, generally above the market interest rate.

Coupon paid without conditions

Coupon independent of underlying's performance

Conditional capital protection through the barrier

As long as no stock touches it, you get the full denomination back, even if the stocks have fallen

Lower risk of loss than a direct investment

Coupon cushion provides a fixed payment you don't have when buying the underlyings

Sources: UBS AG (2018); own illustration

When it suits you, and the risks

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As long as no stock touches it, you get the full denomination back, even if the stocks have fallen

Lower risk of loss than a direct investment
Coupon cushion provides a fixed payment you don't have when buying the underlyings

Risks

Loss possible after a barrier event
after a kick-in you take the worst stock's loss

Limited earnings potential (max. 15%)
You can not earn more than USD 1,150, whatever the stocks do

No entitlement to dividends
Dividends are only paid to shareholders

Issuer risk (UBS)
If the issuer, which is UBS in this case, goes bankrupt, you could lose your money.

Sources: UBS AG (2018); own illustration

The Multi-BRC is a bond minus a put

You lend the issuer 1,000 and sell them a down-and-in put on the worst of three. The coupon is what they pay you for it.

THE PRODUCT

Multi-BRC

Pay USD 1,000 today. Get USD 150 in coupons, then USD 1,000 or shares of the worst stock (final level W).

$$2 \times 75 + 1{,}000 \times \min(1, W)$$

after a kick-in (else USD 1,150)

Sources: Arnold (2026), slide 14 (same payoffs, same price); own illustration

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after a kick-in (else USD 1,150)

=

BOND

UBS bond

Pays the coupons and the USD 1,000, whatever the stocks do.

$$2 \times 75 + 1{,}000$$

in every scenario

Sources: Arnold (2026), slide 14 (same payoffs, same price); own illustration

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UBS bond

Pays the coupons and the USD 1,000, whatever the stocks do.

$$2 \times 75 + 1{,}000$$

in every scenario

−

SOLD PUT

Down-and-in put on the worst of three

You sold it: after a kick-in, you bear the loss.

$$1{,}000 \times \max(0, 1 - W)$$

only after a kick-in

The law of one price
Same payoffs, same price

Sources: Arnold (2026), slide 14 (same payoffs, same price); own illustration

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only after a kick-in

The law of one price
Same payoffs, same price

Bond: Easy to price
just discount it

Put: Hard to price
barrier + three stocks

→ **that's where Monte Carlo comes in**

Sources: Arnold (2026), slide 14 (same payoffs, same price); own illustration

Recap: Where Black-Scholes can be applied

For a vanilla call or put on one stock, there is an exact formula

Lecture reference: **Lecture 4, slide 7**

BLACK-SCHOLES(-MERTON) FORMULA FOR EUROPEAN OPTIONS

CALL

$$C_0 = S_0 N(d_1) - K e^{-rT} N(d_2)$$

PUT

$$P_0 = K e^{-rT}\left(1 - N(d_2)\right) - S_0\left(1 - N(d_1)\right)$$

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

WHERE

C_0, P_0	call / put value today	r	risk-free rate (continuous)
S_0	share price today	T	time to maturity in years
K	strike price	σ	volatility per year
$N(\cdot)$	cumulative standard normal distribution		

Sources: Arnold (2026), slide 7; Black & Scholes (1973)

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K	strike price	σ	volatility per year
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WHAT THE MODEL ASSUMES

1. **One stock**
   a single underlying
2. **Only the final price matters**
   the payoff ignores what happened along the way
3. **Constant volatility and rate**
   σ and r stay fixed all year

Sources: Arnold (2026), slide 7; Black & Scholes (1973)

Why Monte Carlo is used for the Worst-Of Multi-BRC

A barrier alone has a formula. Three stocks alone have a formula. Both together do not.

SCEN ONE POSSIBLE YEAR · WORST OF ADBE / AAPL / MSFT — VOL 45/28/32% · CORR 0.14–0.41 · OUR EXCEL'S INPUTS



TRADING DAY	ADBE	AAPL	MSFT	KICK-IN	PAYOUT PER USD 1,000
0	100.0%	100.0%	100.0%	–	–

Sources: Arnold (2026), slides 9–10; Rubinstein & Reiner (1991); Stulz (1982); Johnson (1987); own simulation

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TRADING DAY	ADBE	AAPL	MSFT	KICK-IN	PAYOUT PER USD 1,000
252	87.9%	114.8%	110.1%	–	–

Sources: Arnold (2026), slides 9–10; Rubinstein & Reiner (1991); Stulz (1982); Johnson (1987); own simulation

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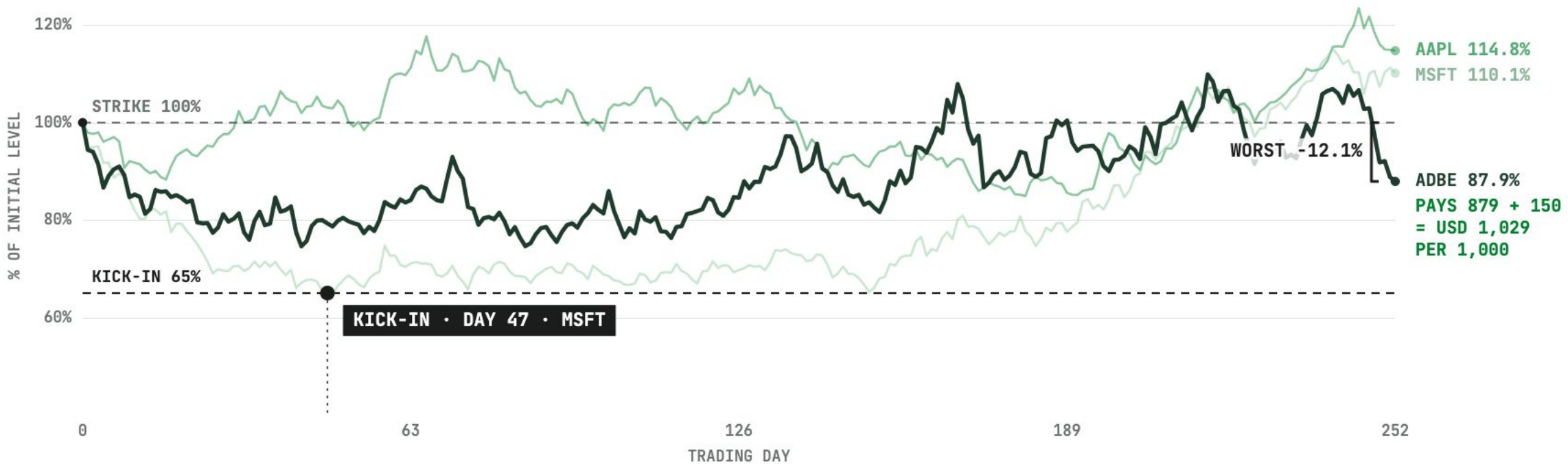
TRADING DAY	ADBE	AAPL	MSFT	KICK-IN	PAYOUT PER USD 1,000
252	87.9%	114.8%	110.1%	DAY 47 · MSFT	–

Sources: Arnold (2026), slides 9–10; Rubinstein & Reiner (1991); Stulz (1982); Johnson (1987); own simulation

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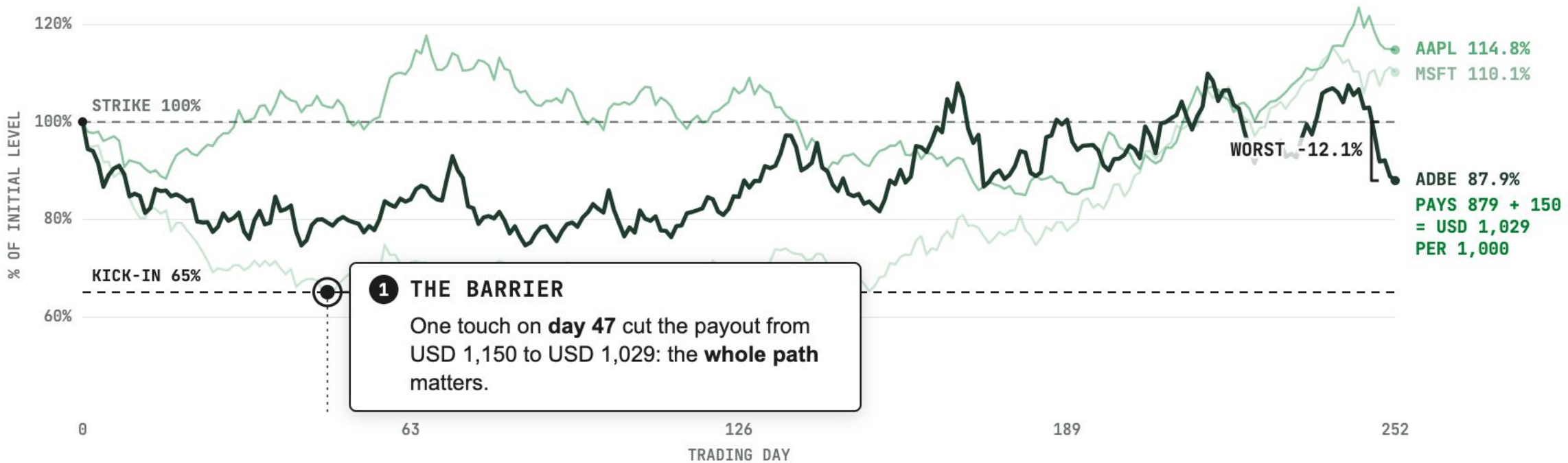
TRADING DAY	ADBE	AAPL	MSFT	KICK-IN	PAYOUT PER USD 1,000
252	87.9%	114.8%	110.1%	DAY 47 · MSFT	879 + 150 = USD 1,029

Sources: Arnold (2026), slides 9–10; Rubinstein & Reiner (1991); Stulz (1982); Johnson (1987); own simulation

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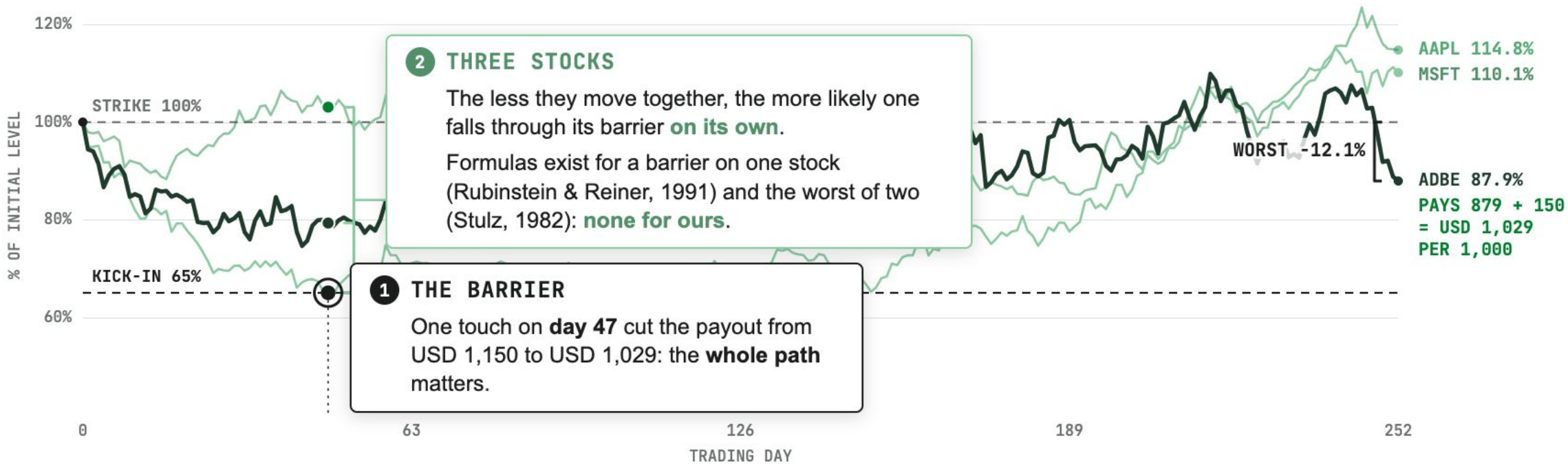
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SCEN ONE POSSIBLE YEAR · WORST OF ADBE / AAPL / MSFT — VOL 45/28/32% · CORR 0.14–0.41 · OUR EXCEL'S INPUTS



2 THREE STOCKS

The less they move together, the more likely one falls through its barrier **on its own**.

Formulas exist for a barrier on one stock (Rubinstein & Reiner, 1991) and the worst of two (Stulz, 1982): **none for ours**.

1 THE BARRIER

One touch on **day 47** cut the payout from USD 1,150 to USD 1,029: the **whole path** matters.

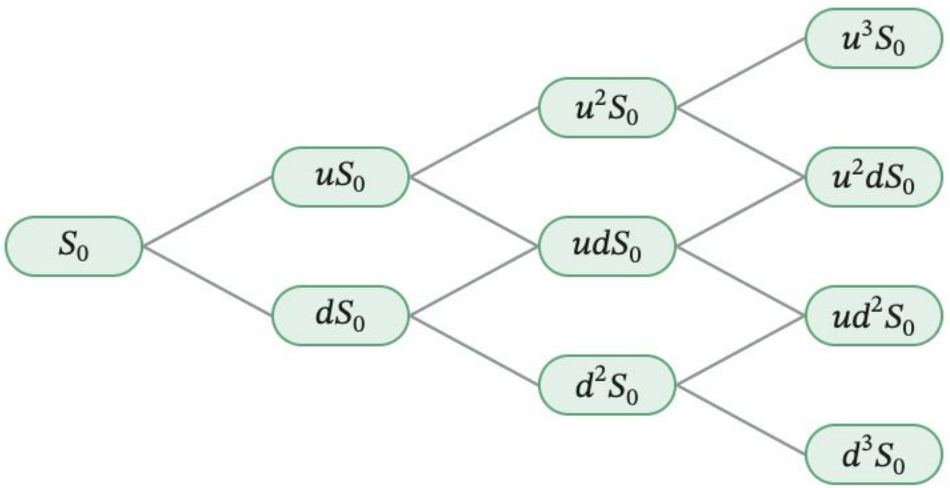
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Why Monte Carlo is used for the Worst-Of Multi-BRC

A barrier alone has a formula. Three stocks alone have a formula. Both together do not.

3 The binomial tree



1 stock, 3 steps: 10 nodes

3 stocks × 252 trading days

≈ 1.03 billion nodes

$$\sum_{k=0}^{252} (k+1)^3 = 1{,}032{,}401{,}161$$

Far too big for Excel: the **curse of dimensionality**.

→ So we simulate: Monte Carlo

Sources: Arnold (2026), slides 9–10; Rubinstein & Reiner (1991); Stulz (1982); Johnson (1987); own simulation

Pricing the Multi-BRC in seven steps

Our plan for the Excel model, from the market data to the fair value

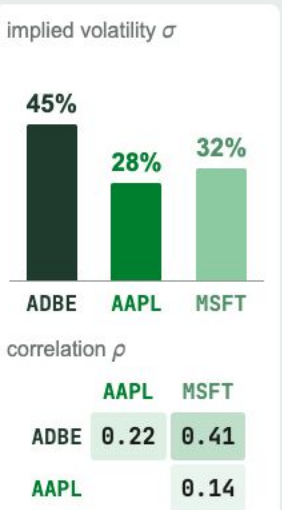
THE PLAN

$$\text{fair value} = \text{bond} - \text{put}$$ | only the **put** needs a simulation

1 The missing inputs

3 4 5 6

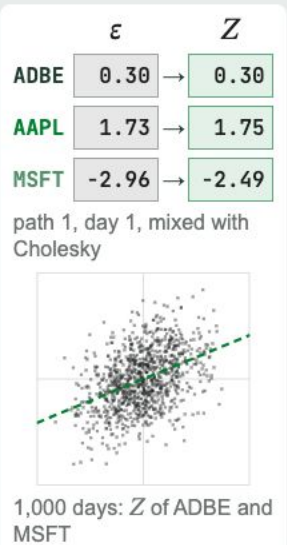
volatility and correlation



2 How do we simulate a stock?

7 4 8

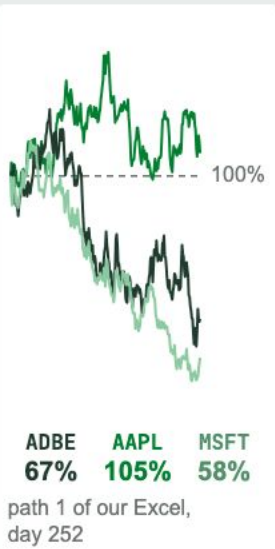
random numbers and correlated shocks



3 The engine, one simulated year

8

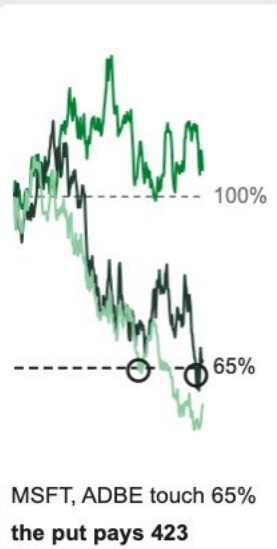
day by day: 252 trading days, 3 stocks



4 Barrier and payoff

8

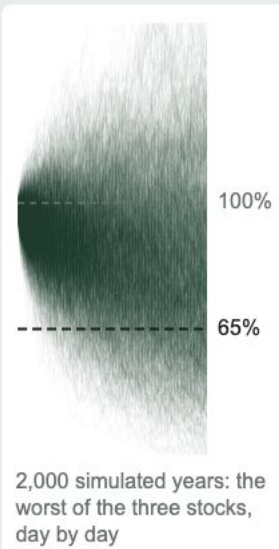
did a stock touch 65%? what does the put pay?



5 2,000 paths with a data table

9

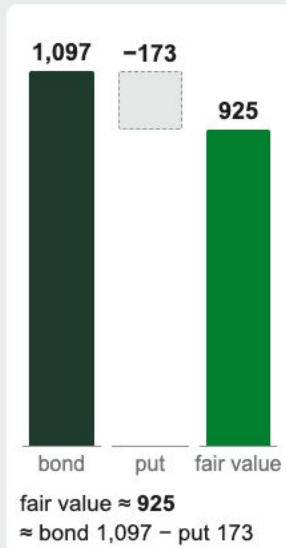
one engine, run once for every path



6 The fair value

10 11

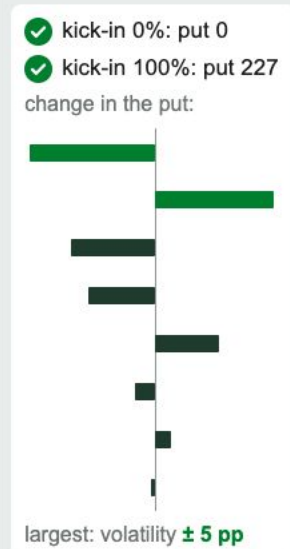
bond − average put, with a 95% interval



7 Checks and sensitivities

11 9

two extreme barriers · which input matters most



Sources: own Excel model; Arnold (2026), slide 15; own simulation

Step 1: the missing inputs

Black-Scholes needs five inputs, and we know four of them

Lecture reference: **Lecture 4, slide 7**

BLACK-SCHOLES(-MERTON) FORMULA FOR EUROPEAN OPTIONS

CALL

$$C_0 = S_0 N(d_1) - K e^{-rT} N(d_2)$$

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$$d_2 = d_1 - \sigma\sqrt{T}$$

THE FIVE INPUTS

S_0	stock price today, USD
K	strike
T	maturity
r	risk-free rate
σ	volatility

Sources: Arnold (2026), slide 7; Black & Scholes (1973); Bloomberg (2026); own Excel model, sheets 2, 5 and 6

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THE FIVE INPUTS

S_0	stock price today, USD ADBE 238.33 · AAPL 332.80 · MSFT 534.39	KNOWN
K	strike **each quoted option's strike** (sheet 5)	KNOWN
T	maturity **≈ 0.95 years:** the listed expiry closest to redemption (sheet 5)	KNOWN
r	risk-free rate **4.43%** · USD 1-year risk-free rate (Bloomberg, 6 Oct 2026)	KNOWN
σ	volatility	

Sources: Arnold (2026), slide 7; Black & Scholes (1973); Bloomberg (2026); own Excel model, sheets 2, 5 and 6

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$$d_2 = d_1 - \sigma\sqrt{T}$$

option price → **Black-Scholes backwards** → σ implied volatility · 50 options per stock (sheet 6)

Black-Scholes can't price our product, but it finds the missing piece.

THE FIVE INPUTS

S_0	stock price today, USD ADBE 238.33 · AAPL 332.80 · MSFT 534.39	KNOWN
K	strike **each quoted option's strike** (sheet 5)	KNOWN
T	maturity **≈ 0.95 years:** the listed expiry closest to redemption (sheet 5)	KNOWN
r	risk-free rate **4.43%** · USD 1-year risk-free rate (Bloomberg, 6 Oct 2026)	KNOWN
σ	volatility **?** from option prices (sheets 5 and 6)	MISSING

Sources: Arnold (2026), slide 7; Black & Scholes (1973); Bloomberg (2026); own Excel model, sheets 2, 5 and 6

Step 2: How do we simulate a stock?

The lecture's geometric Brownian motion, one small time step at a time

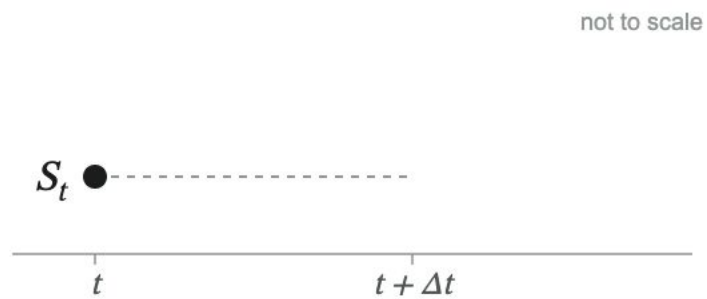
Lecture reference: **Lecture 4, slides 11 and 15**

GBM IN DISCRETE TIME STEPS

$$S_{t+\Delta t} = S_t + \text{Drift} + \text{Shock}$$

$\Delta t = T / N = 1 / 252$: one trading day

ONE TIME STEP



1 Drift (risk-neutral)

2 Random shock

3 $Z \sim \mathcal{N}(0, 1)$

Sources: Arnold (2026), slides 11 and 15; own Excel model, sheet 7; own illustration

Step 2: How do we simulate a stock?

The lecture's geometric Brownian motion, one small time step at a time

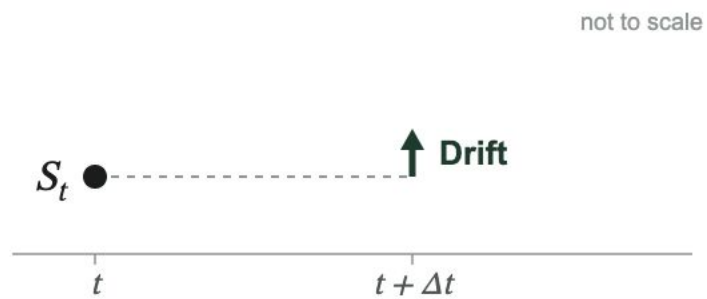
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$$S_{t+\Delta t} = S_t + \textbf{Drift} + \textbf{Shock}$$

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ONE TIME STEP



1 Drift (risk-neutral)

$$(r - q) \cdot S_t \cdot \Delta t$$

lecture: r_f and δ

risk-free rate minus dividend yield:
risk-neutral, no expected returns needed

our Excel: r = 4.43%,
q = ADBE 0 · AAPL 0.32 · MSFT 0.73%

2 Random shock

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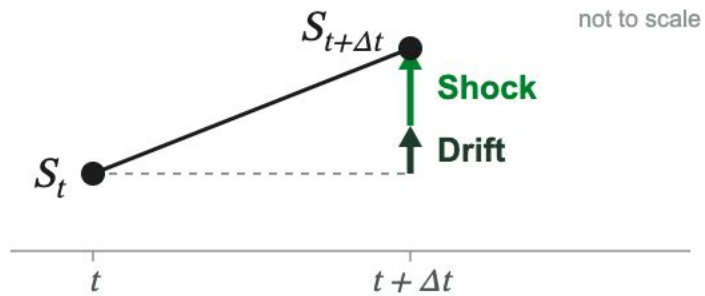
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2 Random shock

$$\sigma \cdot S_t \cdot \sqrt{\Delta t} \cdot Z$$

its size depends on the **volatility** σ

our Excel (step 1):
σ = ADBE 45 · AAPL 28 · MSFT 32%
$\sqrt{\Delta t}$: the variance grows with time

3 $Z \sim \mathcal{N}(0,1)$

Sources: Arnold (2026), slides 11 and 15; own Excel model, sheet 7; own illustration

Step 2: How do we simulate a stock?

The lecture's geometric Brownian motion, one small time step at a time

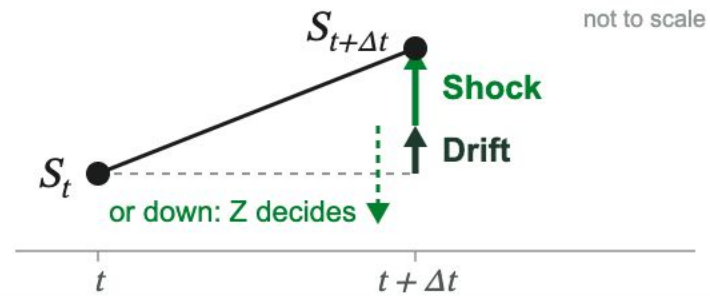
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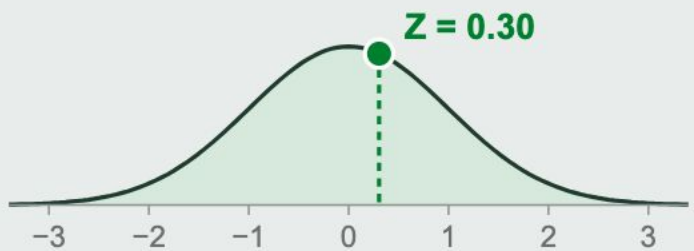
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$\sqrt{\Delta t}$: the variance grows with time

3 $Z \sim \mathcal{N}(0,1)$

a random number from a **bell curve**



path 1, day 1, Adobe: Z = 0.3047

One Z per stock × day × path: 3 × 252 × 2,000 ≈ 1.5 million | `=NORM.S.INV(RAND())` → Paste as values | "So the prices don't change every time Excel recalculates."

Sources: Arnold (2026), slides 11 and 15; own Excel model, sheet 7; own illustration

Step 2: correlated shocks

Our stocks don't move independently, so we mix the random numbers

$\varepsilon_1, \varepsilon_2, \varepsilon_3$: three **independent** random numbers a day, from sheet 7

CHOLESKY: MIX THE RANDOM NUMBERS

$$Z_{\text{ADBE}} = \varepsilon_1$$

keeps its own random number

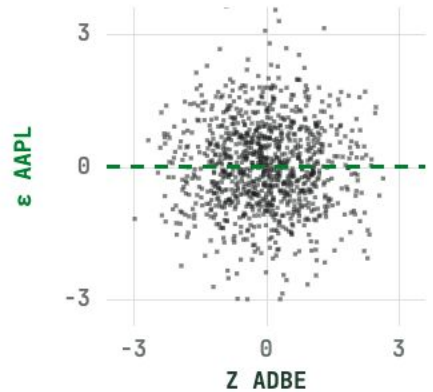
$$Z_{\text{AAPL}} = 0.22 \cdot \varepsilon_1 + 0.98 \cdot \varepsilon_2$$

$$Z_{\text{MSFT}} = 0.41 \cdot \varepsilon_1 + 0.05 \cdot \varepsilon_2 + 0.91 \cdot \varepsilon_3$$

Notation adapted from Hull (2022), ch. 21, who writes x for the independent numbers and ε for the correlated ones.

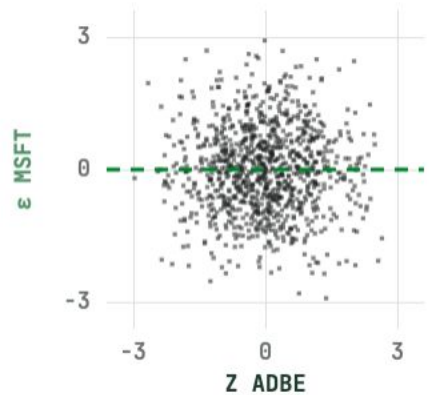
EXCEL · SHEET 4, STEP D · SHEET 8, H49

CORR 1,000 SIMULATED DAYS — OWN SIMULATION



ADBE vs AAPL
INDEPENDENT

CORRELATION
0.00
TARGET 0.00



ADBE vs MSFT
INDEPENDENT

CORRELATION
-0.01
TARGET 0.00

Sources: Glasserman (2003), ch. 2–3; Hull (2022), ch. 21; own Excel model, sheets 4 and 8; own simulation

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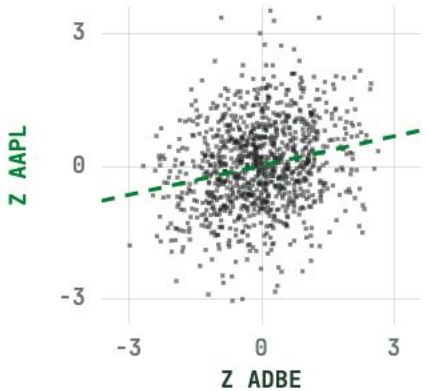
0.22 = their correlation · **0.98** = $\sqrt{1 - 0.22^2}$

$$Z_{\text{MSFT}} = 0.41 \cdot \varepsilon_1 + 0.05 \cdot \varepsilon_2 + 0.91 \cdot \varepsilon_3$$

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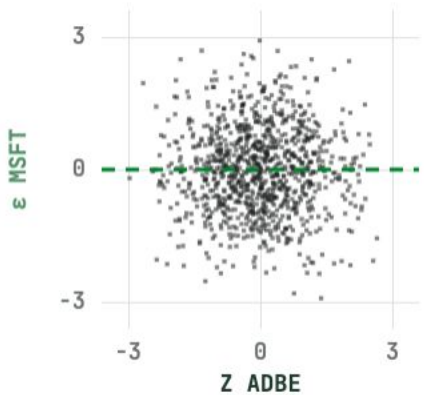
EXCEL · SHEET 4, STEP D · SHEET 8, H49

CORR 1,000 SIMULATED DAYS — OWN SIMULATION



ADBE vs AAPL

CORRELATION
0.22
TARGET 0.22



ADBE vs MSFT
INDEPENDENT

CORRELATION
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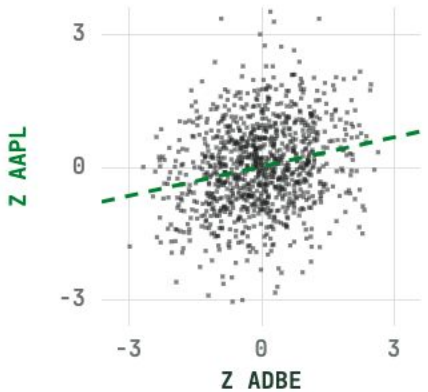
$$Z_{\text{MSFT}} = 0.41 \cdot \varepsilon_1 + 0.05 \cdot \varepsilon_2 + 0.91 \cdot \varepsilon_3$$

mixes all three: correlation **0.41** with Adobe, **0.14** with Apple

Notation adapted from Hull (2022), ch. 21, who writes x for the independent numbers and ε for the correlated ones.

EXCEL · SHEET 4, STEP D · SHEET 8, H49

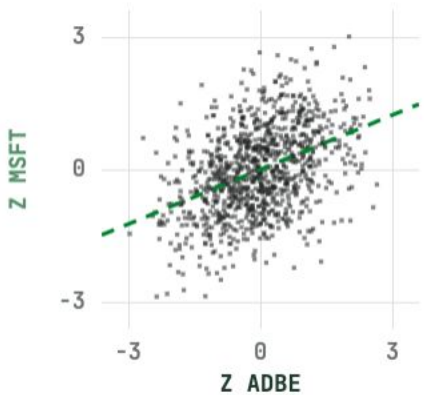
CORR 1,000 SIMULATED DAYS — OWN SIMULATION



ADBE vs AAPL

CORRELATION **0.22**

TARGET 0.22



ADBE vs MSFT

CORRELATION **0.41**

TARGET 0.41

AAPL vs MSFT · CORRELATION **0.14** · TARGET 0.14

Sources: Glasserman (2003), ch. 2–3; Hull (2022), ch. 21; own Excel model, sheets 4 and 8; own simulation

Step 3: the engine, one simulated year.

Sheet 8 copies the step down 252 trading days, for all three stocks

Lecture reference: **Lecture 4, slide 15**

THE ENGINE'S FORMULA

$$S(t) = S(t-1) \times \mathrm{EXP}\left(\left(r - q - \sigma^2 / 2\right) \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z \right)$$

EXCEL · SHEET 8, CELL K49

Z:

EXP:

PATH PATH 1 · 252 TRADING DAYS · % OF EACH STOCK'S START — OUR EXCEL, SHEET 8



Sources: Arnold (2026), slide 15; own Excel model, sheet 8 (path 1)

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EXCEL · SHEET 8, CELL K49

Z: the correlated shock from step 2

EXP:

PATH PATH 1 · 252 TRADING DAYS · % OF EACH STOCK'S START — OUR EXCEL, SHEET 8



A QUIET YEAR | MONTE CARLO | THE WORST-OF MULTI-BRC | WHY MONTE CARLO | **PRICING IN EXCEL** | CONCLUSION

Sources: Arnold (2026), slide 15; own Excel model, sheet 8 (path 1)

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EXCEL · SHEET 8, CELL K49

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EXP: a price can never become negative
$-\sigma^2 / 2$: half the variance, so the average still grows at $r - q$

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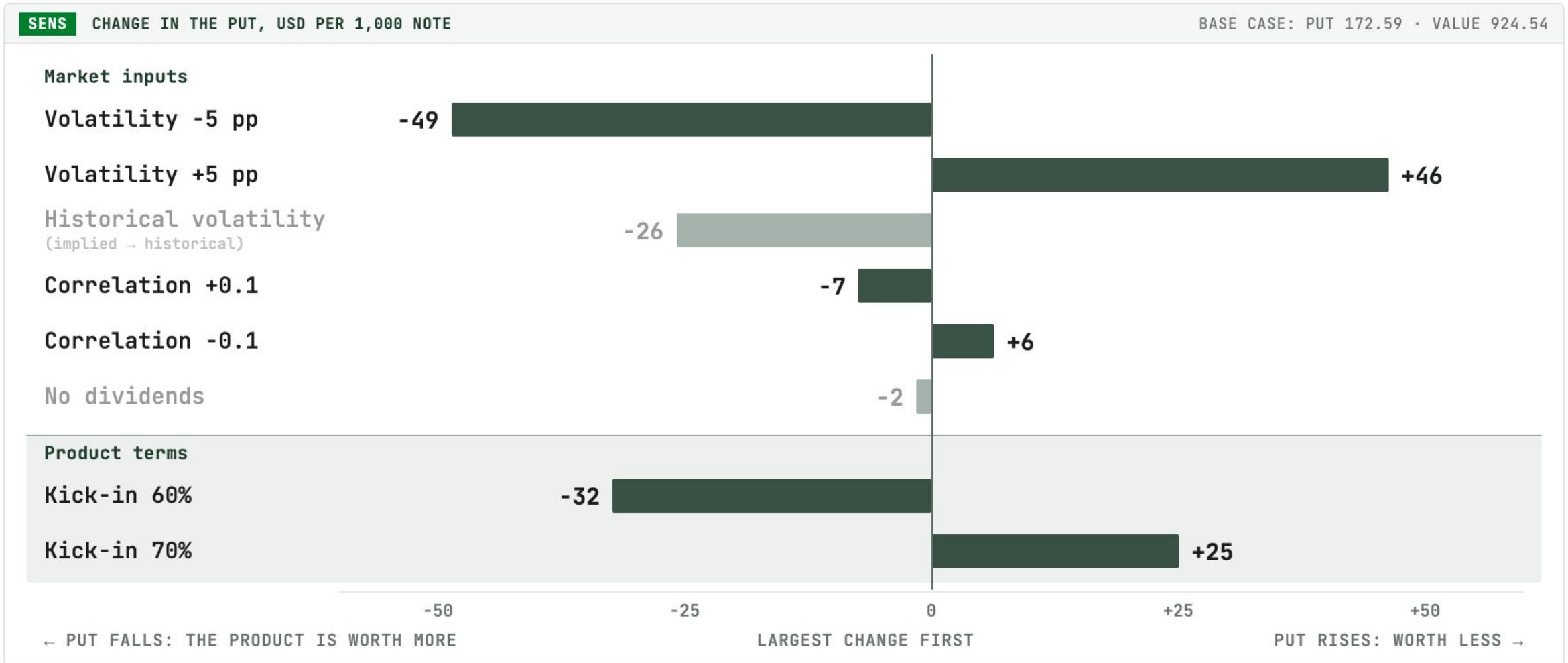
PATH PATH 1 · 252 TRADING DAYS · % OF EACH STOCK'S START — OUR EXCEL, SHEET 8



Sources: Arnold (2026), slide 15; own Excel model, sheet 8 (path 1)

Which input matters most?

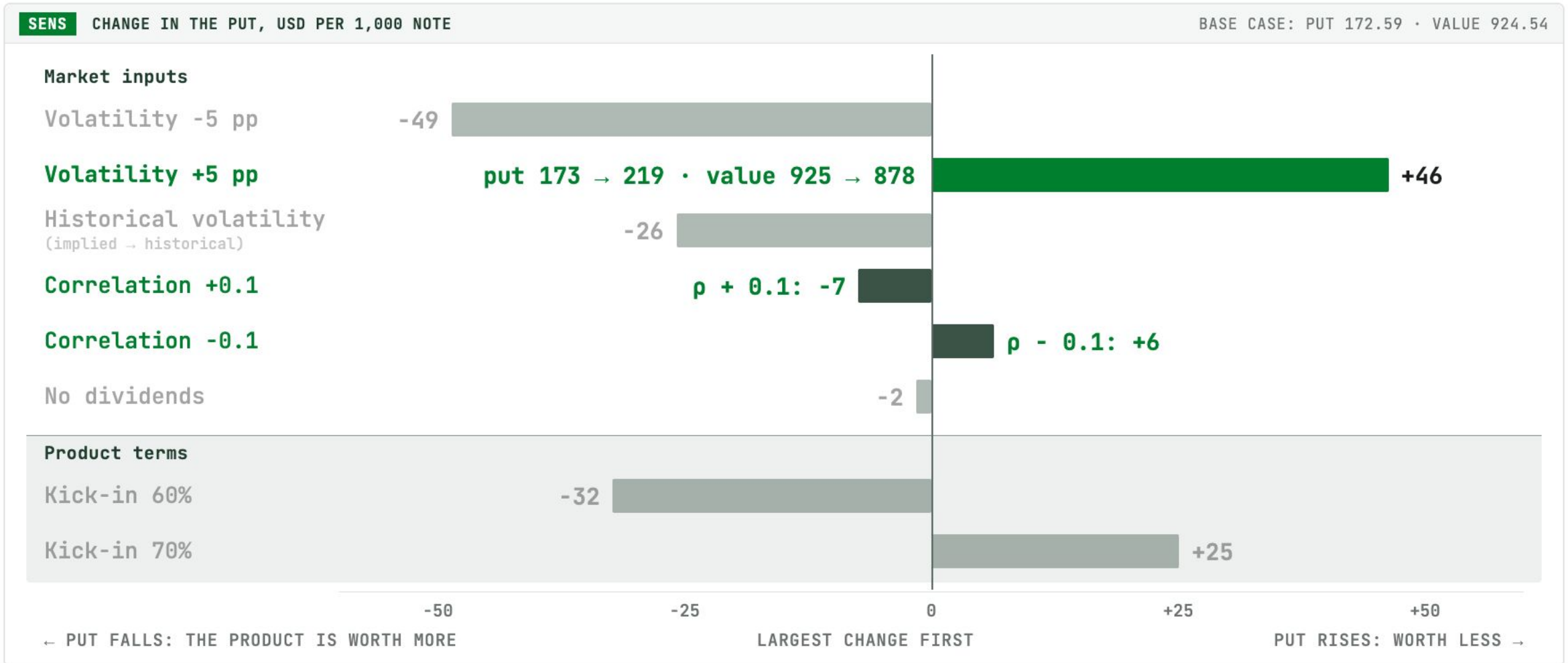
Sheet 11: how much the put changes when one input changes (same 2,000 paths)



Sources: own Excel model, sheet 11

Which input matters most?

Sheet 11: how much the put changes when one input changes (same 2,000 paths)



Sources: own Excel model, sheet 11

Summary of the video

The whole video in four lines

1 **Monte Carlo**
Simulate many futures, average the discounted payoffs



$$\text{price} \approx e^{-rT} \cdot \frac{1}{n} \sum_{i=1}^{n} \text{payoff}_i$$

Sources: Arnold (2026), slides 7, 10 and 15; own Excel model

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2 **The Multi-BRC**
A bond minus a put on the worst of three stocks

Bond
coupons + 1,000

−

Put
on the worst of
ADBE **AAPL** **MSFT**

Sources: Arnold (2026), slides 7, 10 and 15; own Excel model

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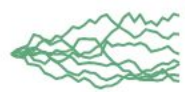
–

Put
on the worst of
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3 **Why Monte Carlo**
Black-Scholes can't price it, a binomial tree is far too big

$N(d_1)$ ✗ **Black-Scholes**

 ✗ **Binomial tree**

 ✓ **Monte Carlo**

Sources: Arnold (2026), slides 7, 10 and 15; own Excel model

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Bond coupons + 1,000 **–** **Put** on the worst of **ADBE AAPL MSFT**

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Black-Scholes can't price it, a binomial tree is far too big



4 **The Excel recipe**

Inputs SHEETS 1–6 → **Random shocks** SHEET 7 → **Paths** SHEET 8 → **Payoff** SHEETS 8–9 → **Average** SHEET 10

Sources: Arnold (2026), slides 7, 10 and 15; own Excel model

Limits and next steps

Three limits of our model, and what fixes each one.

LIMITATION

1

Daily barrier check

Once a day, not continuously



→

NEXT STEP

Barrier correction

Broadie et al. (1997)

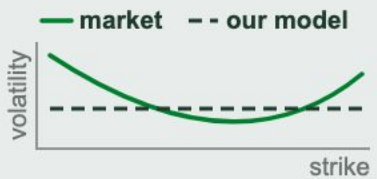
$$65\% \times e^{+0.5826\sigma\sqrt{\Delta t}} \approx 66.1\%$$

our daily 65% acts like a continuous 63.9%;
to price the continuous 65%,
check daily against ≈ 66.1%
(Adobe, σ = 45%, Δt = 1/252)

Sources: Broadie et al. (1997); own Excel model, Read Me and sheet 11; own illustration

Limits and next steps

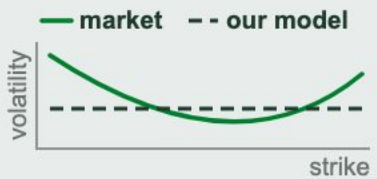
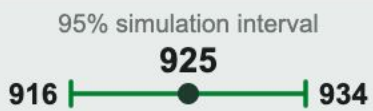
Three limits of our model, and what fixes each one.

LIMITATION		NEXT STEP
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2	**One flat volatility** One σ per stock, no smile 	→ **Stochastic volatility** Volatility moves at random, too 

Sources: Broadie et al. (1997); own Excel model, Read Me and sheet 11; own illustration

Limits and next steps

Three limits of our model, and what fixes each one.

	LIMITATION		NEXT STEP
1	**Daily barrier check** Once a day, not continuously		**Barrier correction** Broadie et al. (1997) $65\% \times e^{+0.5826\sigma\sqrt{\Delta t}} \approx 66.1\%$ our daily 65% acts like a continuous 63.9%; to price the continuous 65%, check daily against ≈ 66.1% (Adobe, σ = 45%, Δt = 1/252)
2	**One flat volatility** One σ per stock, no smile		**Stochastic volatility** Volatility moves at random, too	
3	**Correlation from past data** One year of daily returns	ADBE · AAPL ρ = 0.22 ADBE · MSFT ρ = 0.41 AAPL · MSFT ρ = 0.14	**Stress test (sheet 11)** Fair value with correlation ± 0.1	
+	**Simulation error** 2,000 paths; 925 is an estimate		**More paths** 4× the paths halve the interval	

Two more limitations are in the workbook's Read Me: the discounting choice and the option maturity gap.

Sources: Broadie et al. (1997); own Excel model, Read Me and sheet 11; own illustration

Is 15% enough?

Who gets what: fair coupon vs. what you get

Lecture reference: **Lecture 4, slide 11**



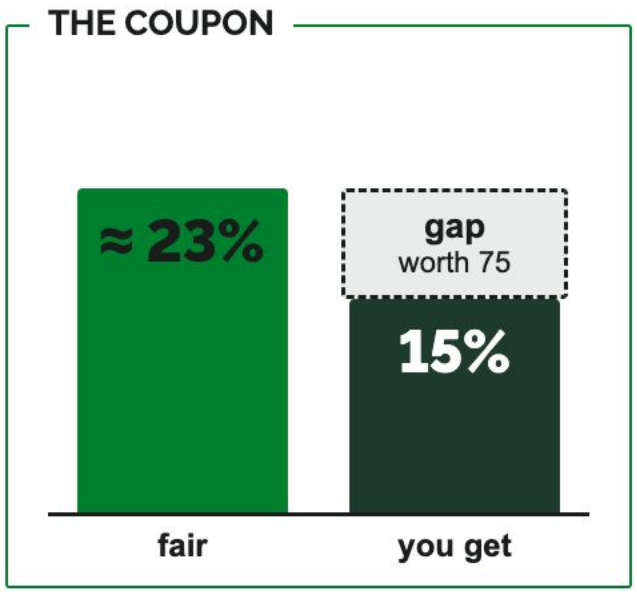
Fair: the coupon that would make it worth 1,000

Sources: own Excel model, sheets 10 and 11; own count over the 2,000 paths of sheet 9 (891 years; sheet 10, C27, 46.85%, counts losses before coupons); Arnold (2026), slide 11

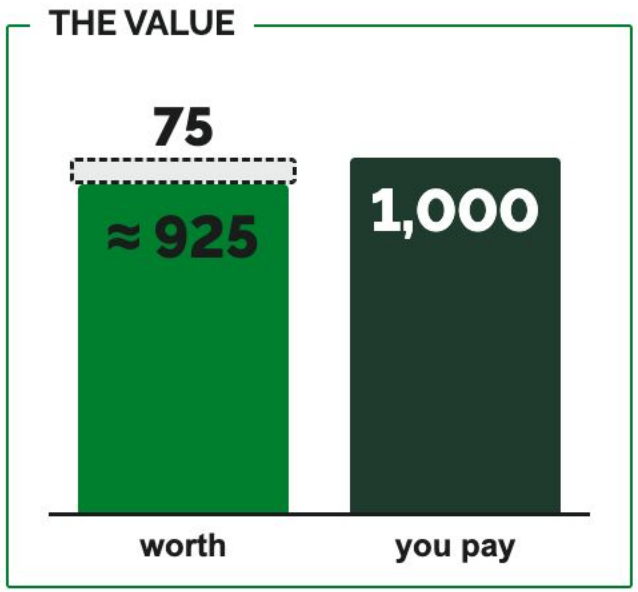
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Who gets what: fair coupon vs. what you get

Lecture reference: **Lecture 4, slide 11**



Fair: the coupon that would make it worth 1,000



The bank keeps **75** (we assume: hedging, risk, margin)

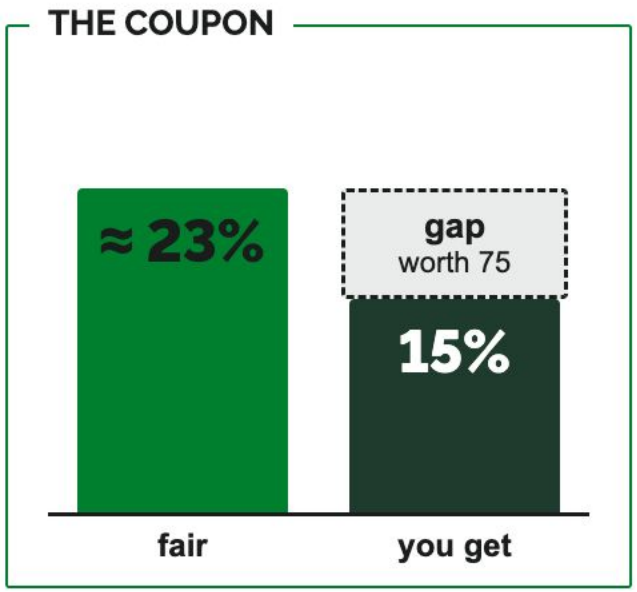
Volatility −5 pts → bank keeps **≈3%**
Today → **≈7.5%** (75 of 1,000)
Volatility +5 pts → bank keeps **≈12%**

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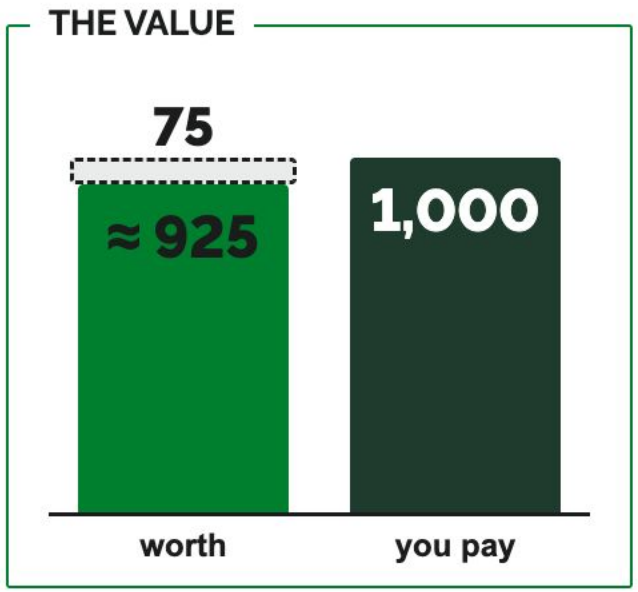
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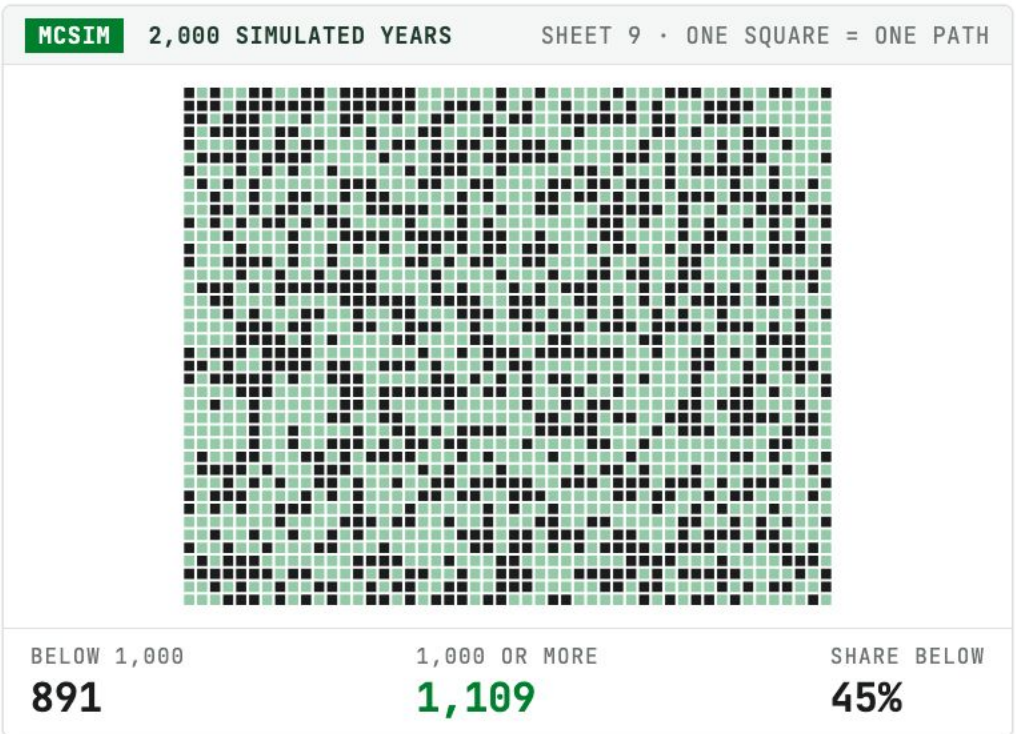


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Volatility +5 pts → bank keeps **≈12%**



45% of our simulated years end below 1,000, coupons included

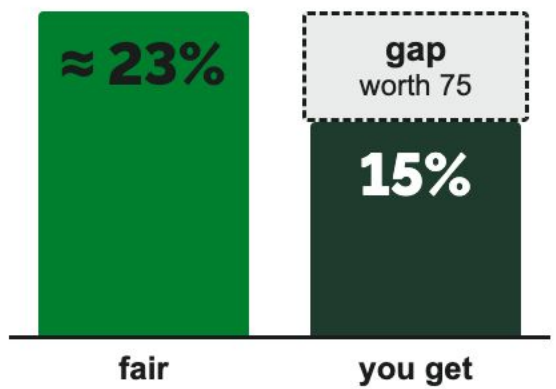
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Is 15% enough?

Who gets what: fair coupon vs. what you get

Lecture reference: **Lecture 4, slide 11**

THE COUPON



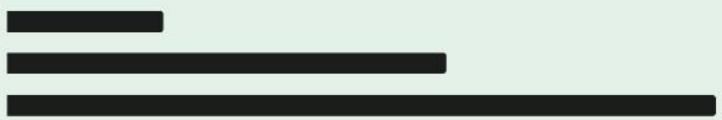
Fair: the coupon that would make it worth 1,000

THE VALUE

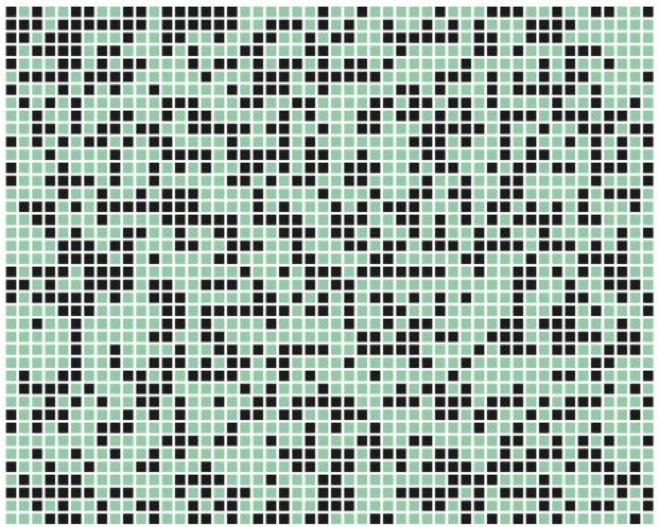


The bank keeps **75** (we assume: hedging, risk, margin)

Volatility −5 pts → bank keeps **≈3%**
Today → **≈7.5%** (75 of 1,000)
Volatility +5 pts → bank keeps **≈12%**



MCSIM 2,000 SIMULATED YEARS — SHEET 9 · ONE SQUARE = ONE PATH



BELOW 1,000	1,000 OR MORE	SHARE BELOW
891	1,109	45%

45% of our simulated years end below 1,000, coupons included

Risk-neutral numbers, not a forecast: if stocks earn more, the real chance of a loss is lower

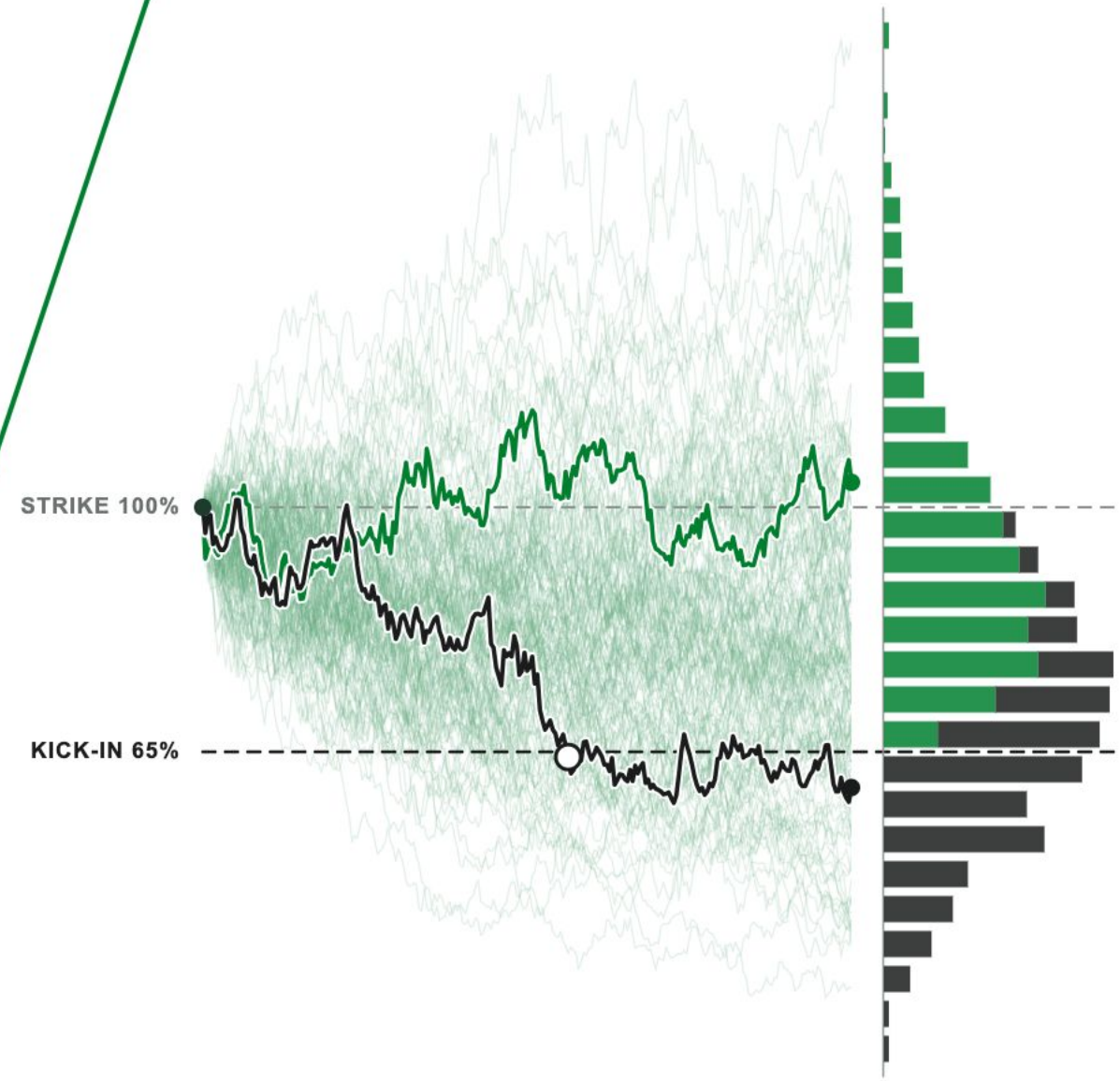
Lecture 4, slide 11

Sources: own Excel model, sheets 10 and 11; own count over the 2,000 paths of sheet 9 (891 years; sheet 10, C27, 46.85%, counts losses before coupons); Arnold (2026), slide 11

CORPORATE FINANCE (7,107)
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Thank you for watching

Next time you see a **15% coupon**, ask yourself what it really costs you.



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Where we use them: Lecture 4 (Arnold, 2026), slides 7 (Black-Scholes), 9–10 (binomial tree), 11–13 (risk-neutral valuation), 14 (replication) and 15 (Monte Carlo and geometric Brownian motion). Hull (2022), chapter 21, and Glasserman (2003), chapters 1–3: Monte Carlo error and correlated random numbers. Stulz (1982) and Johnson (1987): formulas for options on the worst of two stocks and of several stocks. Swiss Structured Products Association (2026): category 1230, barrier reverse convertible. The explanation in section 2 of the video ("What is Monte Carlo?") is adapted from The Synthetic Mind (2026), whose visual style also inspired the web app; all drawings and animations are our own. UBS AG (2018) is the model for our product's structure; our version delivers shares where the 2018 note paid cash.